

# Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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# Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options  $\Rightarrow$  high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs are undervaluing applicants with certain characteristics, future such applicants may receive currently-withheld opportunities

# Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
    - Goal is to accept applicants with highest success probabilities who will matriculate
    - Per literature, measure success with program completion and job placement ranking
    - Predictors of acceptance only (success only) may be overweighted (underweighted)
  - ② If they are making mistakes, how can they make better admissions decisions?
    - In turn, by changing decision rules, how much better of a cohort can they admit?
- ★ This paper's methods generalize to admissions, hiring, and other settings

# Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
  - Correctly weighting these characteristics would increase admitted cohort's average success probability by 9–10 percentage points without reading more files

# Contribution to Literature

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
  - Krueger & Wu (2000); Athey, Katz, Krueger, Levitt & Poterba (2007); Stock, Finegan & Siegfried (2006–2015); Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
  - **Optimality:** Cho & Rust (2010); Anderson, Rosen, Rust & Wong (2025)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

# Synthetic Data Protocol (IRB-Approved)

## ① Administrative office receives data from university departments

- Algorithmically extracts variables from textual files, such as recommendation letters

## ② Synthetic data (Patki et al. 2016) used for model development

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with  $< 5$  comparable observations binned to prevent attribute disclosure

## ③ Administrative office executes final estimation code on actual data

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

# Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality Test

# Data Description

- **Sample:** Applicants to R1 university's economics PhD program from 2015–2019
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Read File  $\Rightarrow$  Accept  $\Rightarrow$  Matriculate  $\Rightarrow$  Success
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, statements of purpose, CVs, and supplemental forms using dictionaries to construct variables

# Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
  - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Results:** Different independent variables best predict each dependent variable
  - **Accept:** *Committee Score* is best predictor. Other predictors: demographics, *GRE Quantitative Percentile*, *Work Experience*, *CV Coding* (negative)
  - **Matriculate:** Predictors of acceptance often negatively predict matriculation. Best predictor of matriculation is *Open House* attendance
  - **Success:** *PhD Program Rank* is best predictor. Other predictors: *Age*, *TOEFL/IELTS*, *Work Experience*, *Essay Positive Terms (%)*

# Acceptance vs. Success AMEs

| Dependent Variable          | Accept     | T100 Placement |
|-----------------------------|------------|----------------|
| Winsorized Age              | -0.0001    | 0.0053*        |
| Female                      | 0.0163**   | -0.0032        |
| International Asian         | -0.0214*** | -0.0127        |
| GRE Quantitative Percentile | 0.0011**   | 0.0006         |
| TOEFL/IELTS                 | -0.0022    | 0.0409*        |
| Work Experience             | 0.0140**   | 0.0358**       |
| #Professor Recommenders     | -0.0033    | -0.0069        |
| Missing TOEFL/IELTS         | 0.0070     | -0.0008        |
| Committee Score             | 0.0788***  | 0.0007         |
| CV Coding                   | -0.0044*   | 0.0125         |
| Essay Economics Terms (%)   | 0.0001     | -0.0104        |
| Essay Positive Terms (%)    | -0.0013    | 0.0123*        |

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

» Insufficiency of Comparing AMEs

» Regressions vs. Model

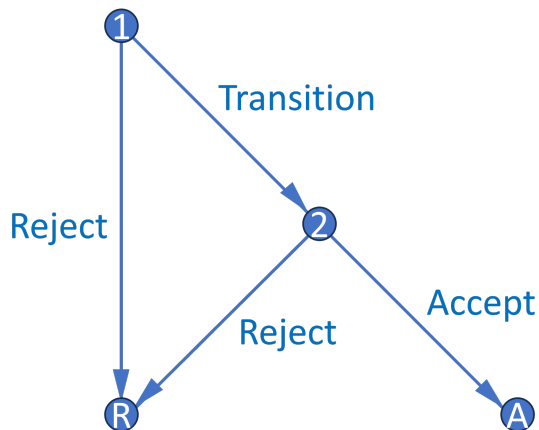
# Outline

- 1 Regression Analysis
- 2 Structural Models**
- 3 Structural Estimation
- 4 Optimality Test

# Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
  - Per literature, committee should accept applicants with highest success probabilities
  - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
  - Quantify gains from deviating from actual to counterfactual decision rule

# Dynamics



- **Round 1:** See only objective variables. Transition to Round 2 or reject
- **Round 2:** See survivors' objective and subjective variables. Accept or reject
- Assume for now number of applicants transitioned and accepted is fixed
- Matriculation model will account for success and matriculation probabilities

# Portfolio Choice Problem

- Accept applicants with highest success probabilities under capacity constraints
- For applicant  $n$ , choose  $d_{1,n} \in \{\text{Transition}, \text{Reject}\}$  and  $d_{2,n} \in \{\text{Accept}, \text{Reject}\}$ :

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A,$$

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

where  $x$  = objective variables,  $z$  = subjective variables,  $y \in \{\text{Success}, \text{Failure}\}$ ,  
 $N = \#$  of applicants, and  $N_T$  and  $N_A = \max \#$  of files read and acceptances

# Portfolio Choice Problem

- Accept applicants with highest success probabilities under capacity constraints

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N_T} \underbrace{\mathbb{1}(d_{2,n} = A)}_{\text{Acceptance}} \leq N_A$$

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N \underbrace{EV_{2,n}(x_n; \bar{\lambda}_2^*)}_{\mathbb{E}[\text{Round 2 Value}]} \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_{1,n} = T)}_{\text{Transition}} \leq N_T,$$

- Prove with Lagrangians that this problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

# Dynamic Model

- **States:**  $r$  = round,  $x$  = objective variables,  $z$  = subjective variables,  $t$  = year,  $\bar{k}$  = fixed program rank,  $\epsilon_r$  = T1EV preference shock with scale parameter  $\sigma$
- **Controls:**  $d_1 \in \{\text{Transition, Reject}\}$ ,  $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:**  $y \in \{\text{Success, Failure}\}$ ,  $y_r^*$  = marginal outcome

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## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

# Dynamic Model

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S|t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\tilde{z}|x, t)$$

# Dynamic Model

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

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$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left( e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\tilde{z}|x, t)$$

## Conditional Choice Probabilities:

$$P(d_1|x, t, \bar{k}) = \frac{\exp(u_1(\cdot, d_1)/\sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(\cdot, \tilde{d}_1)/\sigma)}, \quad P(d_2|x, z, t, \bar{k}) = \frac{\exp(u_2(\cdot, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(\cdot, \tilde{d}_2)/\sigma)}$$

# Portfolio Choice with Matriculation

- Accept applicants with highest success probabilities who are expected to matriculate
- For applicant  $n \in \{1, \dots, N\}$ , choose  $d_n \in \{\text{Accept}, \text{Reject}\}$  given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where  $(x, z)$  = variables,  $y \in \{\text{Success}, \text{Failure}\}$ ,  $m \in \{\text{Matriculate}, \text{Decline}\}$ ,  
 $N$  = number of applicants, and  $N_M$  = maximum number of matriculants

# Portfolio Choice with Matriculation

- Accept applicants with highest success probabilities who are expected to matriculate

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M$$

## Theorem 1 (Proof)

For large  $N$ , this problem has a cutoff rule solution on **success** probabilities.

- Likely matriculants more valuable, but also more expensive  $\Rightarrow$  effects cancel
- Matriculation probabilities matter in Round 1 of a two-round model

# Matriculation-Dynamic Model

- **States:**  $r$  = round,  $x$  = objective variables,  $z$  = subjective variables,  $t$  = year,  $\bar{k}$  = fixed program rank,  $\epsilon_r$  = T1EV preference shock with scale parameter  $\sigma$
- **Controls:**  $d_1 \in \{\text{Transition, Reject}\}$ ,  $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:**  $y \in \{\text{Success, Failure}\}$ ,  $m \in \{\text{Matriculate, Decline}\}$ ,  $y_r^*, m_1^*$  = cutoffs

# Matriculation-Dynamic Model

- **States:**  $r$  = round,  $x$  = objective variables,  $z$  = subjective variables,  $t$  = year,  $\bar{k}$  = fixed program rank,  $\epsilon_r$  = T1EV preference shock with scale parameter  $\sigma$
- **Controls:**  $d_1 \in \{\text{Transition, Reject}\}$ ,  $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:**  $y \in \{\text{Success, Failure}\}$ ,  $m \in \{\text{Matriculate, Decline}\}$ ,  $y_r^*, m_1^* = \text{cutoffs}$

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

# Matriculation-Dynamic Model

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Round 2 Value}]} + \underbrace{L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}$$

# Matriculation-Dynamic Model

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

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## Importance of Matriculation:

$$\lim_{\sigma \rightarrow 0} EV_2 = \mathbb{E}_{\tilde{z}}[\max\{P(y = S | x, \tilde{z}, t, \bar{k}) - P(y_2^* = S | t), 0\} P(m = M | x, \tilde{z}, t) | x, t, \bar{k}]$$

- Matriculation matters when expected success probability  $\gg$  Round 2 cutoff
- **Upshot:** Low-ranked programs must consider matriculation in Round 1

# Outline

- 1 Regression Analysis
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# Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$
$$\prod_{n=1}^N \underbrace{P_{d_{1,n}}(\theta_T)^{\mathbb{1}(d_{1,n}=T)} (1 - P_{d_{1,n}}(\theta_T))^{\mathbb{1}(d_{1,n}=R)}}_{\text{Round 1 conditional choice probabilities}} \prod_{n=1}^{N_T} \underbrace{P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Round 2 conditional choice probabilities}}$$

- In unrestricted likelihood  $\mathcal{L}^U$ , parameterize CCPs with flexible logistic specifications
- In restricted likelihood  $\mathcal{L}^R$ , use model CCPs, which depend on structural parameters
- $\mathcal{L}^R$  aligns success probabilities implied by outcomes with those implied by CCPs

# Cutoff Probability Identification

## Theorem 2 (Proof)

*Cutoff probability estimate is such that number of acceptances  $N_A$  equals expected  $N_A$ . For any cutoff and  $N_A$ , likelihood is highest when committee accepts best  $N_A$  applicants.*

- Identify each year's cutoffs by how selective committee is that year
  - More acceptances and stronger applicants increase cutoff
  - Fewer acceptances and weaker applicants decrease cutoff
- While selectivity identifies cutoffs, sorting improves restricted likelihood

# Restricted Dynamic Estimates

|       | R1 Cutoff Level            | R2 Cutoff Probability      | Sigma              |
|-------|----------------------------|----------------------------|--------------------|
| 2015  | 0.3731<br>(0.3067, 0.4539) | 36.58%<br>(29.81%, 43.92%) |                    |
| 2016  | 0.3795<br>(0.3025, 0.4761) | 37.21%<br>(29.20%, 45.99%) |                    |
| 2017  | 0.4126<br>(0.3563, 0.4778) | 40.27%<br>(34.52%, 46.30%) |                    |
| 2018  | 0.3836<br>(0.3271, 0.4497) | 37.68%<br>(31.79%, 43.97%) |                    |
| 2019  | 0.3887<br>(0.3268, 0.4624) | 38.14%<br>(31.89%, 44.81%) |                    |
| Sigma |                            |                            | 0.0020<br>(0.0013) |

# Restricted Matriculation-Dynamic Estimates

|       | R1 Cutoff Level            | R2 Cutoff Probability      | Sigma                |
|-------|----------------------------|----------------------------|----------------------|
| 2015  | 0.0071<br>(0.0026, 0.0194) | 36.35%<br>(29.47%, 43.84%) |                      |
| 2016  | 0.0071<br>(0.0028, 0.0180) | 38.25%<br>(31.05%, 46.00%) |                      |
| 2017  | 0.0105<br>(0.0041, 0.0267) | 40.65%<br>(35.15%, 46.39%) |                      |
| 2018  | 0.0095<br>(0.0038, 0.0240) | 36.59%<br>(30.01%, 43.71%) |                      |
| 2019  | 0.0095<br>(0.0035, 0.0255) | 37.41%<br>(31.04%, 44.26%) |                      |
| Sigma |                            |                            | 0.0020**<br>(0.0010) |

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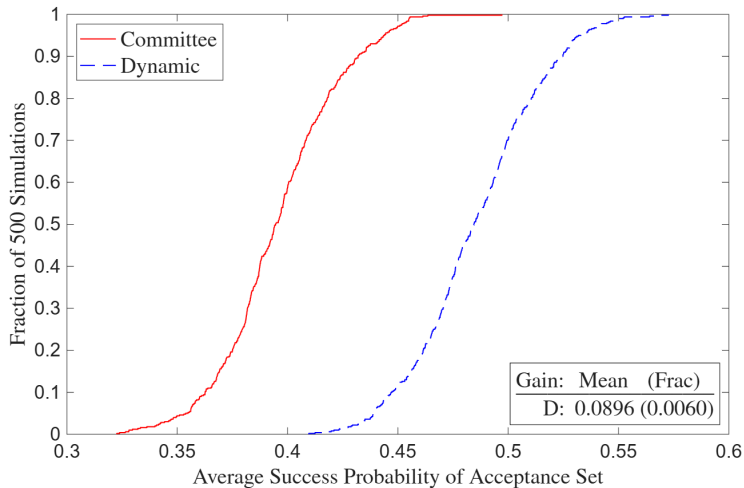
# Deviation Gains Test

- Suppose committee selected model's acceptance/matriculation set. Would average success probability of committee's set have increased, and if so, by how much?
  - Select applicants with model, compute average yearly success probability of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - Plug  $\hat{\theta}_S^U$  into model because  $\hat{\theta}_S^R$  may be biased to rationalize suboptimal behavior
- **Robustness:** Too few observations for holdout set  $\Rightarrow$  start with randomization test
  - Compute average success probabilities with  $B = 500$  perturbations  $\tilde{\theta}_{S,b}^U$  of  $\hat{\theta}_S^U$

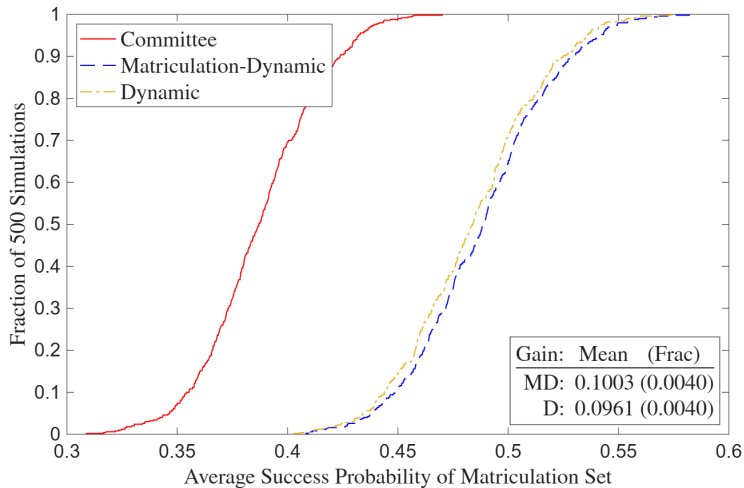
## Theorem 3 (Proof)

*P-value is fraction of perturbations with model's  $\overline{P(\text{Success})} \leq$  committee's*

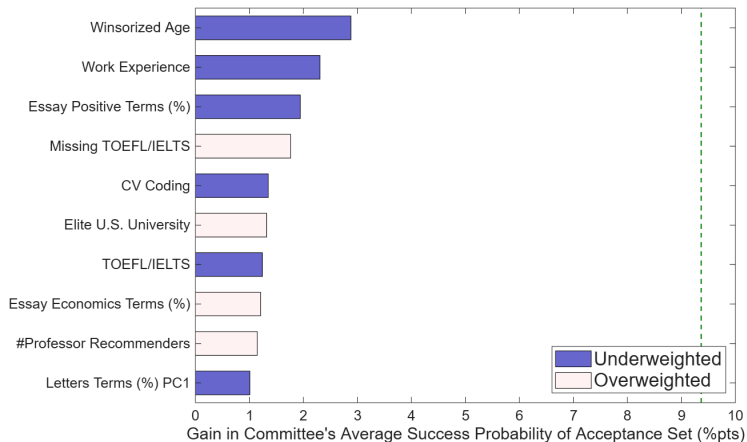
# Deviation Gains (Acceptance)



# Deviation Gains (Matriculation)



# Deviation Gains Attribution (Acceptance)

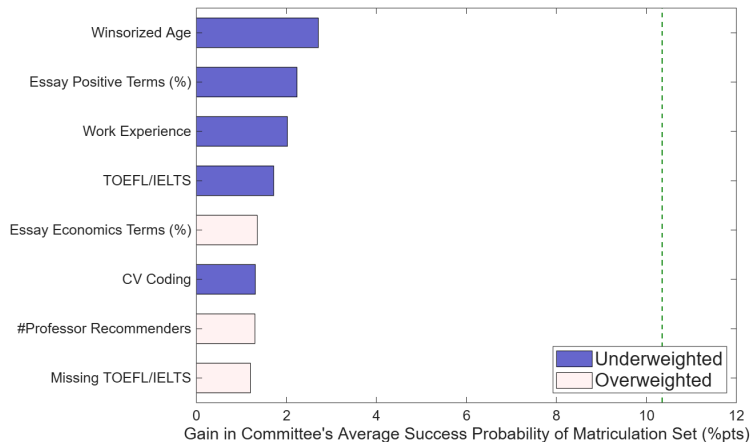


- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Acceptance)

» Deviation Gains Attribution Table (Acceptance)

# Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

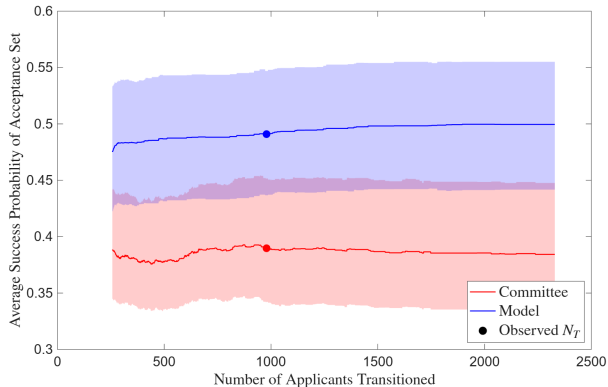
# Regressions vs. Model

| Dependent Variable          | Accept     | T100 Placement |
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| Essay Economics Terms (%)   | 0.0001     | -0.0104        |
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- Blue = underweighted, white = correctly weighted, red = overweighted
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

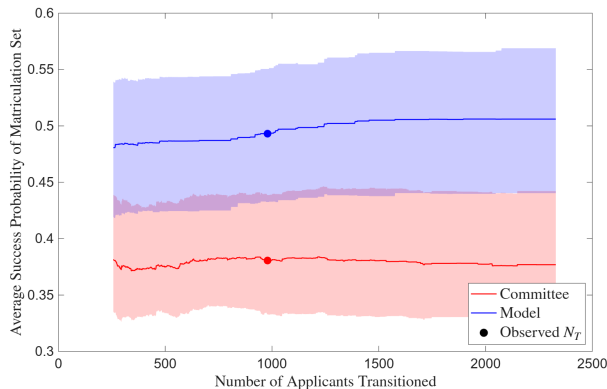
» Acceptance vs. Success AMEs

# Benefit of Reading More Files (Acceptance)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - hwn$
- FOC:  $V = hw / (N_A \bar{P}'(N_T))$
- $V$  is the dollar value of an additional expected success

# Benefit of Reading More Files (Matriculation)



- Suppose committee's curve is increasing and concave
- $\max_{n \in [N_A, N]} VN_A \bar{P}(n) - hwn$
- FOC:  $V = hw / (N_A \bar{P}'(N_T))$
- $V$  is the dollar value of an additional expected success

# Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
  - ① For most applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics are misweighted
  - ② Correctly weighting these characteristics would increase admitted cohort's average success probability by 9–10 percentage points without reading more files

Thank You!

# A1: Synthetic Rows $\neq$ Actual Applicants

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|------------|----------|--------|-------------|---------|-----------------------------|--------|---------------|---------------|---------------|-----------------|--------------|
| 1          | 1        | 0      | 0           | 0       | 89                          | 0      | 0             | 0             | 0             | 5               | 150          |
| 1          | 0        | 1      | 0           | 1       | 96                          | 0      | 1             | 0             | 0             | 6               | 23           |
| 0          | 0        | 0      | 0           | 0       | 88                          | 0      | 1             | 0             | 1             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 80                          | 1      | 1             | 0             | 1             | .               | 150          |
| 0          | 0        | 0      | 0           | 1       | 99                          | 0      | 1             | 0             | 0             | .               | 29           |
| 0          | 0        | 0      | 0           | 0       | 81                          | 0      | 0             | 1             | 1             | .               | 23           |
| 1          | 0        | 1      | 1           | 1       | 92                          | 0      | 1             | 0             | 0             | 3               | 37           |
| 1          | 1        | 1      | 1           | 1       | 93                          | 0      | 0             | 0             | 1             | 3               | 37           |
| 0          | 0        | 0      | 0           | 0       | 83                          | 1      | 0             | 1             | 0             | .               | 48           |
| 1          | 1        | 0      | 0           | 0       | 87                          | 0      | 1             | 0             | 1             | 5               | 34           |



| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|------------|----------|--------|-------------|---------|-----------------------------|--------|---------------|---------------|---------------|-----------------|--------------|
| 0          | 0        | 0      | 0           | 0       | 77                          | 0      | 0             | 0             | 1             | .               | 30           |
| 0          | 0        | 0      | 0           | 1       | 89                          | 0      | 0             | 0             | 0             | .               | 25           |
| 0          | 0        | 0      | 0           | 0       | 85                          | 1      | 0             | 0             | 0             | .               | 150          |
| 0          | 0        | 0      | 0           | 0       | 76                          | 0      | 1             | 0             | 1             | .               | 51           |
| 1          | 1        | 0      | 0           | 0       | 88                          | 0      | 1             | 0             | 0             | 8               | 150          |
| 1          | 0        | 1      | 0           | 0       | 88                          | 0      | 0             | 0             | 1             | 5               | 150          |
| 1          | 1        | 0      | 0           | 0       | 84                          | 0      | 0             | 1             | 0             | 2               | 20           |
| 0          | 0        | 0      | 0           | 0       | 82                          | 0      | 1             | 0             | 0             | .               | 35           |
| 0          | 0        | 0      | 0           | 0       | 70                          | 1      | 0             | 0             | 0             | .               | 150          |
| 1          | 1        | 0      | 0           | 1       | 93                          | 0      | 1             | 0             | 0             | 5               | 57           |

# A1: Dependent Variables

Table 1a: Dependent Variables

| Admission                | Success                        |
|--------------------------|--------------------------------|
| <b>1(Transition)</b>     | <b>1(Complete Program)</b>     |
| <i>1(Accept)</i>         | <i>1(Academic Placement)</i>   |
| <b>1(Waitlist)</b>       | <b>1(Government Placement)</b> |
| <b>1(Open House)</b>     | <b>1(Consulting Placement)</b> |
| <i>1(Matriculate)</i>    | <i>1(Tech Placement)</i>       |
| <b>1(Attend Program)</b> | <b>1(Other Placement)</b>      |
| <b>Program Rank</b>      | <b>1(No Placement)</b>         |
|                          | <b>Placement Rank</b>          |
|                          | <b>1(Top 100 Placement)</b>    |

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

# A1: Objective Variables

Table 1b: Objective Independent Variables

| Demographic                               | Performance                          |
|-------------------------------------------|--------------------------------------|
| <b>Year</b>                               | GRE Verbal Percentile                |
| 1 ( <i>Covid Year</i> )                   | GRE Quantitative Percentile          |
| Winsorized Age                            | GRE Analytic Percentile              |
| 1 (Female)                                | Undergraduate GPA                    |
| 1 ( <b>U.S.</b> )                         | TOEFL/IELTS                          |
| 1 ( <i>U.S. African/Hispanic/Native</i> ) | 1 (Elite U.S. University)            |
| 1 ( <i>U.S. Asian</i> )                   | 1 (Elite U.S. Liberal Arts College)  |
| 1 ( <i>U.S. Other</i> )                   | 1 (Other U.S. University/College)    |
| 1 (International Asian)                   | 1 (Elite Foreign University)         |
| 1 (International Other)                   | 1 (Other Foreign University)         |
|                                           | 1 (Graduate Degree)                  |
|                                           | 1 (Work Experience)                  |
|                                           | #Professor Recommenders              |
|                                           | #Prolific Recommenders               |
|                                           | <i>Concentration Dummy Variables</i> |

# A1: Subjective Variables

Table 1c: Subjective Independent Variables

| Score/Recommendation Letters     | Supplement/CV/Essay             |
|----------------------------------|---------------------------------|
| Committee Score                  | #Math Courses                   |
| Letters Length                   | 1(Advanced Math)                |
| Positive Terms (%) (YS 2012)     | CV Length                       |
| Negative Terms (%) (YS 2012)     | CV Coding                       |
| Standout Terms (%) (BEMS 2022)   | CV 1(Publication/Working Paper) |
| Ability Terms (%) (BEMS 2022)    | Essay Length                    |
| Research Terms (%) (BEMS 2022)   | Essay 1(Specific Professor)     |
| Grindstone Terms (%) (BEMS 2022) | Essay Economics Terms (%)       |
| Teaching Terms (%) (BEMS 2022)   | Essay Research Terms (%)        |
| Communal Terms (%) (BEMS 2022)   | Essay Positive Terms (%)        |
| Agentic Terms (%) (MHM 2009)     | Essay Negative Terms (%)        |
|                                  | Field Dummy Variables           |

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

# A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

# A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Eonic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

# A1: Dictionaries

## ● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

## ● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

# A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend\*, diligen\*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish  $\hat{\beta}$  vs.  $\hat{y}$  problems
  - If GRE predicts success, committee should use it regardless of why

# A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if  $P(\text{Success}|\text{Skill})$  is an S-curve, then  $\hat{\beta}$  will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
  - Let  $y_n^* = x_n\beta + u_n$ ,  $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$ , and  $y_n = d_n y_n^*$
  - **Step 1:** Run probit of  $d$  on  $z$  and compute inverse Mills  $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
  - **Step 2:** Run OLS or heckprobit of  $y$  on  $(x, \hat{\lambda})$  to get  $\hat{\beta}$
  - **Instruments:** Variables that predict matriculation but not the outcome

# A1: Acceptance, Matriculation, and Success AMEs

| Dependent Variable          | Accept     | Matriculate | T100 Placement |
|-----------------------------|------------|-------------|----------------|
| Winsorized Age              | -0.0001    | -0.0223**   | 0.0053*        |
| Female                      | 0.0163**   | -0.0490     | -0.0032        |
| U.S.                        | -0.0013    | 0.1472*     | 0.0192         |
| International Asian         | -0.0214*** | 0.1644**    | -0.0127        |
| GRE Quantitative Percentile | 0.0011**   | -0.0147**   | 0.0006         |
| Undergraduate GPA           | 0.0195     | -0.1026     | -0.1096**      |
| TOEFL/IELTS                 | -0.0022    | -0.0631     | 0.0409*        |
| Work Experience             | 0.0140**   | 0.1591**    | 0.0358**       |
| Committee Score             | 0.0788***  | -0.0371     | 0.0007         |
| Advanced Math               | 0.0246     | -0.1950**   | -0.0287        |
| Letters Terms (%) PC1       | 0.0055     | 0.0671**    | 0.0159         |
| CV Coding                   | -0.0044*   | -0.0414**   | 0.0125         |
| Essay Economics Terms (%)   | 0.0001     | 0.0366**    | -0.0104        |
| Essay Positive Terms (%)    | -0.0013    | -0.0072     | 0.0123*        |
| Open House                  |            | 0.2615***   |                |
| Program Rank                |            |             | -0.0011***     |
| Missing Program Rank        |            |             | -0.3965***     |

Year Dummies  
 GRE Verbal Percentile  
 GRE Analytic Percentile  
 Elite U.S. University  
 Elite U.S. Liberal Arts College  
 Elite Foreign University  
 Other Foreign University  
 Graduate Degree  
 #Professor Recommenders  
 #Prolific Recommenders  
 #Math Courses  
 Letters Length/100  
 Letters Terms (%) PC2  
 CV Length/100  
 CV Publication/Working Paper  
 Essay Length/100  
 Essay Specific Professor  
 Essay Research Terms (%)  
 Essay Negative Terms (%)  
 Missing Dummies

# A1: Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Work experience has large effect on success, graduate degree smaller effect
  - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
  - **Acceptance Regression:**  $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
  - **Success Regression:**  $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

# A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
  - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
  - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality

# A1: Logistic Wald Test Across Programs

Assume  $(Y^E \perp\!\!\!\perp Y^{NE})|X$ . Let  $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$ ,

and  $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$ . Finally, let  $N = N_E + N_{NE}$ . Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let  $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$ . Then:  
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$ . Compare AMEs per Mood (2010)

# A1: Linear Wald Test Across Programs

Assume  $(Y^E \perp\!\!\!\perp Y^{NE})|X$ , i.e. (Economics  $\perp\!\!\!\perp$  Non-Economics) $|$ Predictors.

Let  $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$ , and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$ . Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let  $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$ . (Can also do likelihood-ratio test.)  
Then  $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e.  $\forall n, \sigma_n^2 = \sigma^2$ )

# A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAs

## A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
  - ① **Static Model:** In Round 1, rank applicants based only on objective variables
  - ② **Myopic Model:** Like dynamic, but distribution of  $z$  not conditional on  $x$
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

## A2: Description of Models

- **Round 2:** If accept applicant, get success probability  $P(y = S|x, z, t, \bar{k})$ , where Success =  $\mathbb{1}(\text{Complete Program})$  or  $\mathbb{1}(\text{Good Placement})$ . If reject applicant, get  $P(y_2^* = S|t) = P(\text{Success})$  of year  $t$ 's marginal accepted applicant
- **Round 1:** If reject applicant, get  $L/P(y_1^* = S|t) = L/P(\text{Success})$  of year  $t$ 's marginal transitioned applicant. Can be less or greater than  $P(y_2^* = S|t)$  depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$  = expected value of Round 2, where  $z$  is integrated out conditional on  $(x, t)$  based on  $F(z|x, t)$
  - **Static:** If transition applicant, get  $P(y = S|x, t, \bar{k})$ , or  $P(\text{Success})$  unconditional on  $z$ . It may differ from  $P(y = S|x, z, t, \bar{k})$ , though it still accounts for  $z$ 's effect on  $y$
  - **Myopic:** In  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ , replace  $F(z|x, t)$  with  $F(z|t)$

## A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

### Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

**Upshot:**  $\lambda^*$  is cutoff probability. Denote it hereafter as  $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

## A2.1: Portfolio Choice to Cutoff Rule with Shocks

### Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

### Theorem 4 (Proof)

For large  $N$ , there is a cutoff rule solution with logistic CCPs.

► Portfolio Choice Problem

## A2: Portfolio Choice to Cutoff Rule, Round 2

### Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

**Upshot:**  $\lambda_2^*$  is cutoff probability. Denote it hereafter as  $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

## A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

**Upshot:**  $\lambda_1^*$  is cutoff level. Denote it hereafter as  $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

## A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

**Upshot:**  $\lambda_1^*$  is cutoff probability. Denote it hereafter as  $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

# A2: Dynamic Parameterization 1

## Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

## Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

## Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

## A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left( \sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[ \cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:**  $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[ \frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:**  $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:**  $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

## A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success mostly unobservable for applicants from fewer than six years ago

## A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

- **Motivation:** Interpretable cutoff probabilities in both rounds

» Dynamic Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

» Dynamic Model

# Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S | x, t, \bar{k}) \mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

# Static Model

- **Motivation:** Interpretable cutoff probabilities in both rounds

## Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

## Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

- $P(y = S|x, t, \bar{k})$  estimates inconsistent, but fitted values account for  $z$ 's effect on  $y$

## A2: Static Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success mostly unobservable for applicants from fewer than six years ago

## A2: Cutoff Rule with Matriculation

### Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda(N_M - \overline{N_M})$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

## A2: Cutoff Rule with Matriculation (Large $N$ )

### Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))] P(m = M|w)f(w)dw + \lambda(N_M - \overline{N}_M)$$

### Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

**Upshot:** Point-identified cutoff probability  $\lambda^*$  is with respect to success probabilities

## A2: Cutoff Rule with Matriculation, Round 2

### Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

## A2: Cutoff Rule with Matriculation, Round 1

### Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

### Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

# A2: Matriculation-Dynamic Likelihood

## Matriculation Probability:

$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

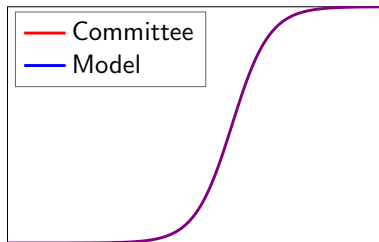
$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}} \\ & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\ & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \end{aligned}$$

## A2: Empirical Considerations

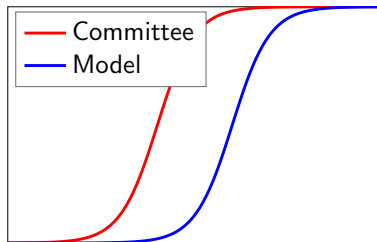
- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in  $z$  as a proxy for this information

## A2: How Deviation Gains Test Works

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

# A2: Deviation Gains Algorithm

## Dynamic/Static/Myopic:

- 1 Rank each  $t$ 's applicants by  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$  with  $\sigma = 0$  for dynamic and myopic, or  $P(y = S|x, t, \bar{k})$  for static. Transition top  $N_T$ , and reject the rest
- 2 Rank surviving applicants by  $P(y = S|x, z, t, \bar{k})$ , and accept top  $N_A$  survivors.  
Compute  $\overline{P(y = S|x, z, t, \bar{k})}$  of acceptance set, and compare to committee's set

## Matriculation-Dynamic/Static:

- 1 Rank each  $t$ 's applicants by  $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$  with  $\sigma = 0$  for dynamic, or  $P(y = S|x, t, \bar{k})$  for static (best analogue). Transition top  $N_T$ , and reject the rest
- 2 Rank surviving applicants by  $P(y = S|x, z, t, \bar{k})$  and accept them in descending order, each matriculating with probability  $P(m = M|x, z, t)$ , until  $N_M$  matriculate.  
Compute  $\overline{P(y = S|x, z, t, \bar{k})}$  of matriculation set, and compare to committee's set

## A2: Randomization Test

### Theorem 5 (Proof)

Let  $d^R$  be the model's decision rule, and  $d^U$  the actual decision rule. By optimality:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ \geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- But for perturbation  $\tilde{\theta}_{S,b}^U$  of  $\hat{\theta}_S^U$ 's asymptotic distribution, can have:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- For matriculation models, compare matriculants, not acceptances

## A2: Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Randomization test is not a true out-of-sample test, and it cannot account for changes in the determinants of success over time, such as writing essays with AI
- Committee may use success-predicting variables that I do not observe
  - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
  - Comprehensive data, especially committee score in  $z$ , protects against this limitation

## A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
  - **Dynamic/Static:** Deviation gains larger for simulated dataset where committee misweights applicant characteristics than for near-optimal simulated dataset
  - **Matriculation-Dynamic/Static:** Same as above
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A

## A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
  - **Example:** Applicants with work experience more likely to succeed and matriculate
  - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes  $P(\text{Success})$ 
  - Instead maximize combination of  $P(\text{Success})$  and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields):  
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$ , where  $\alpha \in [0, 1]$  can be estimated
- **Extension 3:** Waitlist and multiyear models (see online appendix)
  - Three-round models, and two-round models that repeat over infinite horizon
  - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

# A3: Economics Department Summary Statistics

Table 2: Economics Department Summary Statistics

## (a) Objective Variables

|                                   | Full Sample |         | Transition = 1 |         | Accept = 1 |         | Matriculate = 1 |         |
|-----------------------------------|-------------|---------|----------------|---------|------------|---------|-----------------|---------|
|                                   | Mean        | SD      | Mean           | SD      | Mean       | SD      | Mean            | SD      |
| Transition                        | 0.4204      | 0.4937  | 1.0000         | 0.0000  | 1.0000     | 0.0000  | 1.0000          | 0.0000  |
| Waitlist                          | 0.3324      | 0.4712  | 0.7686         | 0.4220  | 0.0743     | 0.2628  | 0.0896          | 0.2877  |
| Accept                            | 0.1103      | 0.3134  | 0.2625         | 0.4402  | 1.0000     | 0.0000  | 1.0000          | 0.0000  |
| Matriculate                       | 0.0339      | 0.1811  | 0.0807         | 0.2725  | 0.3074     | 0.4623  | 1.0000          | 0.0000  |
| Open House                        | 0.0353      | 0.1847  | 0.0817         | 0.2741  | 0.3267     | 0.4702  | 0.5522          | 0.5010  |
| Attend Program                    | 0.6024      | 0.4895  | 0.6752         | 0.4685  | 0.7704     | 0.4214  | 1.0000          | 0.0000  |
| Complete Program                  | 0.5578      | 0.4968  | 0.6210         | 0.4854  | 0.6654     | 0.4728  | 0.6709          | 0.4729  |
| Top 100 Placement                 | 0.1971      | 0.3979  | 0.2472         | 0.4316  | 0.3074     | 0.4623  | 0.3418          | 0.4773  |
| Program Rank                      | 88.9133     | 57.4823 | 78.5844        | 57.1361 | 63.3828    | 54.1845 | 37.0000         | 0.0000  |
| Missing Program Rank              | 0.4199      | 0.4937  | 0.3422         | 0.4747  | 0.2490     | 0.4333  | 0.0000          | 0.0000  |
| Placement Rank                    | 127.7505    | 34.9306 | 123.0222       | 37.2570 | 117.2696   | 40.5746 | 117.4430        | 37.5728 |
| Missing Placement Rank            | 0.4444      | 0.4970  | 0.3830         | 0.4864  | 0.3346     | 0.4728  | 0.3291          | 0.4729  |
| 2016                              | 0.2052      | 0.4040  | 0.1696         | 0.3754  | 0.1790     | 0.3841  | 0.2025          | 0.4045  |
| 2017                              | 0.2014      | 0.4011  | 0.1788         | 0.3833  | 0.1820     | 0.3973  | 0.1772          | 0.3843  |
| 2018                              | 0.1928      | 0.3946  | 0.2686         | 0.4435  | 0.2062     | 0.4054  | 0.2532          | 0.4376  |
| 2019                              | 0.2027      | 0.4021  | 0.2084         | 0.4064  | 0.2179     | 0.4136  | 0.2152          | 0.4136  |
| Winsorized Age                    | 25.9910     | 3.1479  | 26.2635        | 3.1823  | 26.0817    | 3.0652  | 26.3291         | 3.2806  |
| Female                            | 0.3912      | 0.4881  | 0.4147         | 0.4929  | 0.4397     | 0.4973  | 0.4177          | 0.4963  |
| U.S.                              | 0.2357      | 0.4245  | 0.2380         | 0.4261  | 0.2996     | 0.4590  | 0.3038          | 0.4628  |
| International Asian               | 0.4809      | 0.4997  | 0.5230         | 0.4997  | 0.3424     | 0.4754  | 0.4304          | 0.4983  |
| GRE Verbal Percentile             | 68.0129     | 22.5869 | 76.8421        | 18.3703 | 79.5703    | 18.3047 | 76.9744         | 19.4041 |
| GRE Quantitative Percentile       | 88.4550     | 11.6330 | 93.1362        | 5.4135  | 93.5391    | 4.4614  | 92.3077         | 5.7554  |
| GRE Analytic Percentile           | 50.0613     | 27.5744 | 56.3653        | 26.5466 | 60.3398    | 26.0880 | 57.4615         | 26.6481 |
| Undergraduate GPA                 | 3.6387      | 0.3026  | 3.6645         | 0.3063  | 3.7227     | 0.2610  | 3.7068          | 0.2880  |
| TOEFL IELTS                       | 7.4376      | 0.5587  | 7.5641         | 0.5834  | 7.7011     | 0.6667  | 7.4464          | 0.6137  |
| Elite U.S. University             | 0.1267      | 0.3327  | 0.1736         | 0.3790  | 0.2257     | 0.4188  | 0.1772          | 0.3843  |
| Elite U.S. Liberal Arts College   | 0.0296      | 0.1696  | 0.0347         | 0.1832  | 0.0428     | 0.2028  | 0.0633          | 0.2450  |
| Elite Foreign University          | 0.0386      | 0.1928  | 0.0490         | 0.2160  | 0.0350     | 0.1842  | 0.0253          | 0.1581  |
| Other Foreign University          | 0.4049      | 0.4910  | 0.4188         | 0.4936  | 0.3619     | 0.4815  | 0.4430          | 0.4999  |
| Missing Undergraduate Institution | 0.2203      | 0.4145  | 0.1910         | 0.3933  | 0.1984     | 0.3996  | 0.1266          | 0.3346  |
| Graduate Degree                   | 0.5067      | 0.5001  | 0.5730         | 0.4949  | 0.5136     | 0.5008  | 0.5190          | 0.5028  |
| Work Experience                   | 0.5247      | 0.4995  | 0.5894         | 0.4922  | 0.6770     | 0.4685  | 0.7468          | 0.4376  |
| #Professor Recommenders           | 2.6230      | 0.7371  | 2.6302         | 0.7165  | 2.5992     | 0.7118  | 2.5696          | 0.7104  |
| #Profic Recommenders              | 1.2602      | 1.0686  | 1.3912         | 1.0840  | 1.4825     | 1.0424  | 1.3038          | 1.0664  |
| Missing GRE                       | 0.0335      | 0.1800  | 0.0102         | 0.1006  | 0.0039     | 0.0624  | 0.0127          | 0.1125  |
| Missing Undergraduate GPA         | 0.7853      | 0.4107  | 0.7487         | 0.4340  | 0.6770     | 0.4685  | 0.6835          | 0.4681  |
| Missing TOEFL/IELTS               | 0.6076      | 0.4884  | 0.5935         | 0.4914  | 0.6615     | 0.4741  | 0.6456          | 0.4814  |
| Missing Winsorized Age            | 0.0009      | 0.0293  | 0.0000         | 0.0000  | 0.0000     | 0.0000  | 0.0000          | 0.0000  |
| Observations                      | 2329        |         | 979            |         | 257        |         | 79              |         |

## (b) Subjective Variables

|                              | Full Sample |        | Transition = 1 |        | Accept = 1 |        | Matriculate = 1 |        |
|------------------------------|-------------|--------|----------------|--------|------------|--------|-----------------|--------|
|                              | Mean        | SD     | Mean           | SD     | Mean       | SD     | Mean            | SD     |
| Committee Score              | 6.4070      | 1.8701 | 6.4070         | 1.8701 | 8.5003     | 0.8151 | 8.4854          | 0.7157 |
| #Math Courses                | 5.7637      | 1.8154 | 5.8173         | 1.7904 | 5.6992     | 1.8640 | 5.6585          | 1.6219 |
| Advanced Math                | 0.4860      | 0.5004 | 0.4909         | 0.5007 | 0.5680     | 0.4973 | 0.4651          | 0.5047 |
| Letters Length/100           | 8.3474      | 3.5749 | 8.5333         | 3.8263 | 10.0514    | 4.2864 | 9.7088          | 3.4876 |
| Letters Positive Terms (%)   | 11.9770     | 2.2993 | 11.9976        | 2.2077 | 12.2167    | 1.9255 | 12.5110         | 2.0904 |
| Letters Negative Terms (%)   | 5.7286      | 1.3427 | 5.7216         | 1.3318 | 5.7446     | 1.2313 | 5.8938          | 1.2871 |
| Letters Standout Terms (%)   | 0.6241      | 0.4020 | 0.6241         | 0.4029 | 0.7151     | 0.4509 | 0.7077          | 0.5109 |
| Letters Ability Terms (%)    | 1.3326      | 0.6239 | 1.3457         | 0.5964 | 1.3796     | 0.5620 | 1.4334          | 0.6037 |
| Letters Research Terms (%)   | 3.4837      | 1.3319 | 3.5208         | 1.3423 | 3.8548     | 1.1984 | 3.8395          | 1.2031 |
| Letters Grindstone Terms (%) | 1.8189      | 0.7734 | 1.8674         | 0.7808 | 1.9975     | 0.7588 | 2.1600          | 0.8093 |
| Letters Teaching Terms (%)   | 1.7379      | 0.7710 | 1.7489         | 0.7759 | 1.7275     | 0.7536 | 1.8976          | 0.7577 |
| Letters Communal Terms (%)   | 0.5946      | 0.3890 | 0.6019         | 0.3954 | 0.5494     | 0.3234 | 0.5630          | 0.3875 |
| Letters Agentic Terms (%)    | 0.4768      | 0.3150 | 0.4842         | 0.3229 | 0.5039     | 0.3222 | 0.5597          | 0.3314 |
| Letters Terms (%) PC1        | -0.0579     | 1.6183 | 0.0000         | 1.5768 | 0.2176     | 1.3817 | 0.5697          | 1.5386 |
| Letters Terms (%) PC2        | -0.0271     | 1.1177 | 0.0000         | 1.1446 | 0.1002     | 1.1774 | 0.1699          | 1.2823 |
| CV Length/100                | 4.3532      | 2.4237 | 4.3982         | 2.4758 | 4.9641     | 2.7175 | 4.8261          | 2.3473 |
| CV Coding                    | 2.8321      | 2.1135 | 2.7931         | 2.1000 | 2.8618     | 2.1129 | 2.5122          | 1.6450 |
| CV Publication/Working Paper | 0.2518      | 0.4346 | 0.2571         | 0.4377 | 0.3659     | 0.4836 | 0.3902          | 0.4939 |
| Essay Length/100             | 4.5099      | 1.5741 | 4.5173         | 1.5805 | 4.7488     | 1.6373 | 4.9816          | 1.7932 |
| Essay Specific Professor     | 0.4822      | 0.5003 | 0.5219         | 0.5003 | 0.5691     | 0.4972 | 0.6512          | 0.4822 |
| Essay Economics Terms (%)    | 9.7284      | 2.6219 | 9.7216         | 2.6635 | 9.5303     | 2.7324 | 9.8941          | 3.2608 |
| Essay Research Terms (%)     | 5.7975      | 2.0501 | 5.7402         | 2.0168 | 5.7313     | 1.8797 | 5.5231          | 1.8432 |
| Essay Positive Terms (%)     | 11.8293     | 2.6518 | 11.7324        | 2.6650 | 11.4734    | 2.4592 | 11.3169         | 2.4455 |
| Essay Negative Terms (%)     | 6.2844      | 1.9056 | 6.3233         | 1.9712 | 6.2889     | 2.0860 | 6.2651          | 1.9436 |
| Missing Committee Score      | 0.5844      | 0.4929 | 0.0112         | 0.1055 | 0.0195     | 0.1384 | 0.0633          | 0.2450 |
| Missing Application Pdf      | 0.8154      | 0.3881 | 0.6650         | 0.4722 | 0.5136     | 0.5008 | 0.4557          | 0.5012 |
| Missing #Math Courses        | 0.8201      | 0.3842 | 0.6701         | 0.4704 | 0.5214     | 0.5005 | 0.4810          | 0.5028 |
| Missing Letters              | 0.8175      | 0.3863 | 0.6691         | 0.4708 | 0.5136     | 0.5008 | 0.4557          | 0.5012 |
| Missing CV                   | 0.8210      | 0.3835 | 0.6742         | 0.4689 | 0.5214     | 0.5005 | 0.4810          | 0.5028 |
| Missing Essay                | 0.8192      | 0.3849 | 0.6731         | 0.4693 | 0.5214     | 0.5005 | 0.4557          | 0.5012 |
| Observations                 | 2329        |        | 979            |        | 257        |        | 79              |        |

### A3: Economics Department Correlation Matrix

Table 3: Economics Department Correlation Matrix

| Year           | 2000   | 2001   | 2002   | 2003   | 2004   | 2005   | 2006   | 2007   | 2008   | 2009   | 2010   | 2011   | 2012   | 2013   | 2014   | 2015   | 2016   | 2017   | 2018   | 2019   | 2020   | 2021   | 2022   | 2023   | 2024   | 2025   | 2026   | 2027   | 2028   | 2029   | 2030   | 2031   | 2032   | 2033   | 2034   | 2035   | 2036   | 2037   | 2038   | 2039   | 2040   | 2041   | 2042   | 2043   | 2044   | 2045   | 2046   | 2047   | 2048   | 2049   | 2050   | 2051   | 2052   | 2053   | 2054   | 2055   | 2056   | 2057   | 2058   | 2059   | 2060   | 2061   | 2062   | 2063   | 2064   | 2065   | 2066   | 2067   | 2068   | 2069   | 2070   | 2071   | 2072   | 2073   | 2074   | 2075   | 2076   | 2077   | 2078   | 2079   | 2080   | 2081   | 2082   | 2083   | 2084   | 2085   | 2086   | 2087   | 2088   | 2089   | 2090   | 2091   | 2092   | 2093   | 2094   | 2095   | 2096   | 2097   | 2098   | 2099   | 2100   | 2101   | 2102   | 2103   | 2104   | 2105   | 2106   | 2107   | 2108   | 2109   | 2110   | 2111   | 2112   | 2113   | 2114   | 2115   | 2116   | 2117   | 2118   | 2119   | 2120   | 2121   | 2122   | 2123   | 2124   | 2125   | 2126   | 2127   | 2128   | 2129   | 2130   | 2131   | 2132   | 2133   | 2134   | 2135   | 2136   | 2137   | 2138   | 2139   | 2140   | 2141   | 2142   | 2143   | 2144   | 2145   | 2146   | 2147   | 2148   | 2149   | 2150   | 2151   | 2152   | 2153   | 2154   | 2155   | 2156   | 2157   | 2158   | 2159   | 2160   | 2161   | 2162   | 2163   | 2164   | 2165   | 2166   | 2167   | 2168   | 2169   | 2170   | 2171   | 2172   | 2173   | 2174   | 2175   | 2176   | 2177   | 2178   | 2179   | 2180   | 2181   | 2182   | 2183   | 2184   | 2185   | 2186   | 2187   | 2188   | 2189   | 2190   | 2191   | 2192   | 2193   | 2194   | 2195   | 2196   | 2197   | 2198   | 2199   | 2200   | 2201   | 2202   | 2203   | 2204   | 2205   | 2206   | 2207   | 2208   | 2209   | 2210   | 2211   | 2212   | 2213   | 2214   | 2215   | 2216   | 2217   | 2218   | 2219   | 2220   | 2221   | 2222   | 2223   | 2224   | 2225   | 2226   | 2227   | 2228   | 2229   | 2230   | 2231   | 2232   | 2233   | 2234   | 2235   | 2236   | 2237   | 2238   | 2239   | 2240   | 2241   | 2242   | 2243   | 2244   | 2245   | 2246   | 2247   | 2248   | 2249   | 2250   | 2251   | 2252   | 2253   | 2254   | 2255   | 2256   | 2257   | 2258   | 2259   | 2260   | 2261   | 2262   | 2263   | 2264   | 2265   | 2266   | 2267   | 2268   | 2269   | 2270   | 2271   | 2272   | 2273   | 2274   | 2275   | 2276   | 2277   | 2278   | 2279   | 2280   | 2281   | 2282   | 2283   | 2284   | 2285   | 2286   | 2287   | 2288   | 2289   | 2290   | 2291   | 2292   | 2293   | 2294   | 2295   | 2296   | 2297   | 2298   | 2299   | 2300   | 2301   | 2302   | 2303   | 2304   | 2305   | 2306   | 2307   | 2308   | 2309   | 2310   | 2311   | 2312   | 2313   | 2314   | 2315   | 2316   | 2317   | 2318   | 2319   | 2320   | 2321   | 2322   | 2323   | 2324   | 2325   | 2326   | 2327   | 2328   | 2329   | 2330   | 2331   | 2332   | 2333   | 2334   | 2335   | 2336   | 2337   | 2338   | 2339   | 2340   | 2341   | 2342   | 2343   | 2344   | 2345   | 2346   | 2347   | 2348   | 2349   | 2350   | 2351   | 2352   | 2353   | 2354   | 2355   | 2356   | 2357   | 2358   | 2359   | 2360   | 2361   | 2362   | 2363   | 2364   | 2365   | 2366   | 2367   | 2368   | 2369   | 2370   | 2371   | 2372   | 2373   | 2374   | 2375   | 2376   | 2377   | 2378   | 2379   | 2380   | 2381   | 2382   | 2383   | 2384 | 2385 | 2386 | 2387 | 2388 | 2389 | 2390 | 2391 | 2392 | 2393 | 2394 | 2395 | 2396 | 2397 | 2398 | 2399 | 2400 | 2401 | 2402 | 2403 | 2404 | 2405 | 2406 | 2407 | 2408 | 2409 | 2410 | 2411 | 2412 | 2413 | 2414 | 2415 | 2416 | 2417 | 2418 | 2419 | 2420 | 2421 | 2422 | 2423 | 2424 | 2425 | 2426 | 2427 | 2428 | 2429 | 2430 | 2431 | 2432 | 2433 | 2434 | 2435 | 2436 | 2437 | 2438 | 2439 | 2440 | 2441 | 2442 | 2443 | 2444 | 2445 | 2446 | 2447 | 2448 | 2449 | 2450 | 2451 | 2452 | 2453 | 2454 | 2455 | 2456 | 2457 | 2458 | 2459 | 2460 | 2461 | 2462 | 2463 | 2464 | 2465 | 2466 | 2467 | 2468 | 2469 | 2470 | 2471 | 2472 | 2473 | 2474 | 2475 | 2476 | 2477 | 2478 | 2479 | 2480 | 2481 | 2482 | 2483 | 2484 | 2485 | 2486 | 2487 | 2488 | 2489 | 2490 | 2491 | 2492 | 2493 | 2494 | 2495 | 2496 | 2497 | 2498 | 2499 | 2500 |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Transportation | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |

# A3: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

|                        | PC1    | PC2     |
|------------------------|--------|---------|
| Positive               | 0.5044 | -0.1769 |
| Negative               | 0.1479 | 0.3759  |
| Standout               | 0.2115 | -0.5082 |
| Ability                | 0.4667 | -0.0177 |
| Research               | 0.1214 | 0.6418  |
| Grindstone             | 0.3879 | 0.3075  |
| Teaching               | 0.3708 | 0.0743  |
| Communal               | 0.1284 | -0.1983 |
| Agentic                | 0.3768 | -0.1324 |
| Proportion of Variance | 0.2763 | 0.1456  |

# A3: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

| Explanatory Variable              | Intercept           | Gender                | Minorities          | TOEFL Percentile    | TOEFL Percentile    | TOEFL Percentile    | Program Rank        | Program Rank        | Program Rank        |
|-----------------------------------|---------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| 2015                              | -0.3609<br>(0.2665) | -0.2659<br>(0.4761)   | -0.2577<br>(0.4680) | -0.0122<br>(0.1741) | -0.0122<br>(0.1741) | -0.0122<br>(0.1741) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) |
| 2017                              | -0.2664<br>(0.2761) | -0.2347**<br>(0.4712) | -0.1221<br>(0.4673) | -0.0121<br>(0.1737) | -0.0121<br>(0.1737) | -0.0121<br>(0.1737) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) |
| 2019                              | -0.3609<br>(0.2665) | -0.2659<br>(0.4761)   | -0.2577<br>(0.4680) | -0.0122<br>(0.1741) | -0.0122<br>(0.1741) | -0.0122<br>(0.1741) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) | -0.0001<br>(0.1661) |
| Minorized Age                     | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Female                            | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| U.S.                              | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| International Asian               | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| GRE Verbal Percentile             | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| GRE Quantitative Percentile       | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| GRE Reading Percentile            | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| International GRE                 | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| TOEFL/IELTS                       | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Elms U.S. University              | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Elms U.S. Liberal Arts College    | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Elms Foreign University           | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Other Foreign University          | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing Undergraduate Institution | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Graduate Degree                   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Work Experience                   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV/Resume Recommendation          | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV/Resume Recommendation          | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing GRE                       | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing Undergraduate GPA         | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing TOEFL/IELTS               | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Committee Score                   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Admitted Status                   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Letters Length (Y1)               | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Letters Length (Y2)               | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV Length (Y1)                    | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV Length (Y2)                    | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV Coding                         | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| CV Publications/Working Paper     | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Research Grants                   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Research Grants (Y1)              | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Research Grants (Y2)              | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing Committee Score           | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing Application Fee           | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Open Rank                         | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |
| Missing Program Rank              | -0.0010<br>(0.0010) | -0.0010<br>(0.0010)   | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) | -0.0010<br>(0.0010) |

- **Accept ( $p < .1$ ):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ( $p < .1$ ):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

# A3: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

| Explanatory Variable        | Intercept | Female | International Asian | GRE Quantitative Percentile | GRE Analytic Percentile | Work Experience | Committee Score | CV Coding | CV Publishing/Working Paper | Essay Positive Terms (%) | Program Rank | Open House |
|-----------------------------|-----------|--------|---------------------|-----------------------------|-------------------------|-----------------|-----------------|-----------|-----------------------------|--------------------------|--------------|------------|
| Intercept                   | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Female                      | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| International Asian         | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| GRE Quantitative Percentile | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| GRE Analytic Percentile     | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Work Experience             | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Committee Score             | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| CV Coding                   | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| CV Publishing/Working Paper | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Essay Positive Terms (%)    | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Program Rank                | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |
| Open House                  | 0.000     | 0.000  | 0.000               | 0.000                       | 0.000                   | 0.000           | 0.000           | 0.000     | 0.000                       | 0.000                    | 0.000        | 0.000      |

- **Accept** ( $p < .1$ ): 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success** ( $p < .1$ ): Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

## A3: Economics Department Logistic Regression Results (Synthetic)

Table 6a: Economics Department Logistic Regression Results (Synthetic)

[illegible]

- **Accept ( $p < .1$ ):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ( $p < .1$ ):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

### A3: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

[illegible]

- **Accept ( $p < .1$ ):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ( $p < .1$ ):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

# A3: Economics Department Lasso Logistic Regression Results

Table 7a: Economics Department Lasso Logistic Regression Results

| Dependent Variable                | Accept  | Accept  | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|-----------------------------------|---------|---------|-------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 2016                              | -0.1107 |         |             |                |                |                |                |                |                |
| 2017                              |         | -0.4928 |             |                |                |                |                |                |                |
| 2018                              |         | -2.4456 | 0.0186      |                |                |                |                |                |                |
| 2019                              |         | -1.7752 |             |                |                |                |                |                |                |
| Winsorized Age                    |         |         |             | 0.0084         | 0.0047         | 0.0184         |                |                |                |
| Female                            | 0.2674  | 0.5589  |             |                |                |                |                |                |                |
| U.S.                              | -0.1959 |         |             |                |                |                | 0.9477         |                |                |
| International Asian               | -0.8666 | -0.8651 | 0.1271      |                |                |                | -0.5295        | -0.7927        |                |
| GRE Verbal Percentile             | 0.0227  | 0.0047  | -0.0015     | 0.0015         |                |                | -0.0085        |                |                |
| GRE Quantitative Percentile       | 0.1027  | 0.0174  | -0.0273     | 0.0179         | 0.0125         |                | -0.2706        | -0.2243        | -0.0070        |
| GRE Analytic Percentile           | 0.0056  | 0.0035  |             |                |                |                |                |                |                |
| Undergraduate GPA                 | 0.9612  | 0.7135  |             |                |                | -0.1316        |                | 0.2561         |                |
| TOEFL/IELTS                       | 0.3055  |         |             |                |                | 0.0934         | -0.6903        | -0.5850        | -2.2692        |
| Elite U.S. University             | 0.4024  |         | -0.2943     | 0.0300         |                |                |                |                |                |
| Elite U.S. Liberal Arts College   | 0.1112  |         |             |                |                |                |                |                |                |
| Elite Foreign University          | -0.4025 |         |             |                |                |                |                |                |                |
| Other Foreign University          | 0.0790  | 0.0940  | 0.1998      |                |                |                |                |                | -0.2265        |
| Missing Undergraduate Institution |         | -0.0036 |             |                |                |                |                |                |                |
| Graduate Degree                   | 0.1178  |         |             |                |                |                |                |                |                |
| Work Experience                   | 0.6006  | 0.4068  | 0.0404      | 0.0961         | 0.0470         | 0.1976         |                |                | -2.0809        |
| #Professor Recommenders           |         |         |             |                |                |                |                |                | 0.0022         |
| #Prolific Recommenders            | 0.1534  | 0.0301  | -0.0870     | 0.0526         | 0.0363         |                | -0.0789        | -0.0358        |                |
| Missing GRE                       | -0.0534 |         |             | -0.0697        |                |                |                |                |                |
| Missing Undergraduate GPA         | -0.2348 | -0.1148 |             |                |                |                |                |                |                |
| Missing TOEFL/IELTS               | 0.1447  |         |             | -0.1243        |                | -0.1295        | 1.8402         | 2.2548         |                |
| Committee Score                   |         | 3.4243  |             |                |                | 0.0353         |                | -0.4899        |                |
| #Math Courses                     | -0.0553 | -0.0354 |             |                |                |                |                |                |                |
| Advanced Math                     | 0.4302  | 0.7352  |             |                |                |                |                |                |                |
| Letters Length/100                | 0.1196  | 0.0587  |             |                |                |                |                |                |                |
| Letters Terms (%) PC1             | 0.0746  | 0.1485  | 0.1258      |                |                | 0.0102         |                |                |                |
| Letters Terms (%) PC2             | -0.0162 | -0.2001 |             |                |                |                |                |                |                |
| CV Length/100                     | 0.0219  | 0.0298  |             | 0.0014         |                | -0.0286        | -0.1179        |                |                |
| CV Coding                         | -0.0038 | -0.1754 |             | 0.0165         | 0.0076         | -0.2540        | -0.4154        | -0.0998        |                |
| CV Publication/Working Paper      | 0.5563  | 0.6436  |             | 0.0892         |                | -0.0987        |                |                |                |
| Essay Length/100                  | 0.1202  | 0.0295  |             |                |                |                |                |                |                |
| Essay Specific Professor          | 0.3403  | 0.3958  | 0.0674      |                |                |                |                |                |                |
| Essay Economics Terms (%)         | -0.0016 |         |             | -0.0405        | -0.0325        | -0.0104        | 0.4473         | 0.6570         | 0.2117         |
| Essay Research Terms (%)          |         | -0.0278 |             |                |                |                |                | -0.1506        |                |
| Essay Positive Terms (%)          | -0.0049 | -0.0382 |             | 0.0192         | 0.0140         | 0.0267         |                |                |                |
| Essay Negative Terms (%)          |         | -0.0417 |             |                |                |                |                |                |                |
| Missing Committee Score           |         | -0.9416 |             |                | -0.2615        | -0.0714        | 4.0146         | 0.7971         |                |
| Missing Application Pdf           | -1.0365 |         |             |                |                |                |                |                |                |
| Open House                        |         |         | 1.0982      |                |                |                |                |                |                |
| Program Rank                      |         |         |             |                |                | -0.0082        |                |                | 0.1385         |
| Missing Program Rank              |         |         |             |                |                | -2.2557        |                |                | 19.4273        |
| Sample                            | All     | All     | Accept = 1  | All            | All            | All            | All            | All            | All            |
| Observations                      | 2320    | 2320    | 257         | 2320           | 2320           | 2320           | 2320           | 2320           | 2320           |
| Pseudo R <sup>2</sup>             | 0.2841  | 0.3807  | 0.1281      | 0.0216         | 0.0265         | 0.0276         | 0.0276         | 0.2693         |                |
| CV Pseudo R <sup>2</sup>          | 0.2441  | 0.7605  | 0.0596      | 0.0082         | 0.0129         | 0.0199         | 0.0121         | 0.0183         | 0.2638         |

- Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergraduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

## A3: Economics Department Dynamic Estimates

Table 8: Economics Department Dynamic Estimates

|                               | Conjunction Score     |      | Transition            |      | Recede                |      | Summe                 |      |
|-------------------------------|-----------------------|------|-----------------------|------|-----------------------|------|-----------------------|------|
|                               | Coefficient           | RSSE | Coefficient           | RSSE | Coefficient           | RSSE | Coefficient           | RSSE |
| 2016                          | 0.2967***<br>(2.770)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      |
| 2017                          | 0.2967***<br>(2.770)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      |
| 2018                          | 0.2967***<br>(2.770)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      |
| 2019                          | 0.2967***<br>(2.770)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      | 0.2817***<br>(2.417)  |      |
| Mean Age                      | 0.0024***<br>(0.0002) |      | 0.0024***<br>(0.0002) |      | 0.0024***<br>(0.0002) |      | 0.0024***<br>(0.0002) |      |
| Female                        | -0.0043<br>(0.0019)   |      | -0.0043<br>(0.0019)   |      | -0.0043<br>(0.0019)   |      | -0.0043<br>(0.0019)   |      |
| U.S.                          | 0.0017***<br>(0.0001) |      | 0.0017***<br>(0.0001) |      | 0.0017***<br>(0.0001) |      | 0.0017***<br>(0.0001) |      |
| International Res.            | 0.0109***<br>(0.0001) |      | 0.0109***<br>(0.0001) |      | 0.0109***<br>(0.0001) |      | 0.0109***<br>(0.0001) |      |
| Used Virtual Products         | 0.0088***<br>(0.0001) |      | 0.0088***<br>(0.0001) |      | 0.0088***<br>(0.0001) |      | 0.0088***<br>(0.0001) |      |
| QSR Company Quarterly         | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| QSR Analysts Perceptions      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| TQM/ISO Results               | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| U.S. 112 U.S. State College   | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Foreign University            | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Other Foreign University      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Global Integration Initiative | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Working Degree                | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Work Experience               | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Profitability Recommendations | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Using QSR                     | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Using TQM/ISO                 | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| Conjunction Score             | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      | 0.0029***<br>(0.0001) |      |
| (Math Constant)               | -0.0029<br>(0.0001)   |      | -0.0029<br>(0.0001)   |      | -0.0029<br>(0.0001)   |      | -0.0029<br>(0.0001)   |      |
| Advanced Math                 | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| Letters Length                | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| Letters Tens (%) P1           | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P1              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P2              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P3              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P4              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P5              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P6              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P7              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P8              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P9              | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P10             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P11             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P12             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P13             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P14             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P15             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P16             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P17             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P18             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P19             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P20             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P21             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P22             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P23             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P24             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P25             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P26             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P27             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P28             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P29             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P30             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P31             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P32             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P33             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P34             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P35             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P36             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P37             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P38             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P39             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P40             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P41             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P42             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P43             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P44             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P45             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P46             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P47             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P48             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P49             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P50             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P51             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P52             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P53             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P54             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P55             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P56             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P57             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P58             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P59             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P60             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P61             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P62             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P63             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P64             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P65             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P66             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P67             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P68             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P69             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P70             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P71             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P72             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P73             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P74             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P75             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P76             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P77             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P78             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P79             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P80             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P81             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P82             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P83             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P84             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      |
| C Letters (%) P85             | 0.0029<br>(0.0001)    |      | 0.0029<br>(0.0001)    |      | 0.002                 |      |                       |      |

[illegible]

| Obs                                                                   | Form | LL | AE | 2020 | 802 | 2020.8 | 2020.1 |
|-----------------------------------------------------------------------|------|----|----|------|-----|--------|--------|
| Subject standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. |      |    |    |      |     |        |        |

| Obs   Param (LL   $\Delta C$ )            | 2020              | 2021        | 2022        |
|-------------------------------------------|-------------------|-------------|-------------|
| Subsistence level of income in euro/month | 8.0   0.10   0.00 | 8.15   0.11 | 7.90   0.10 |

## A3: Restricted Static Estimates

|                     | R1 Cutoff Probability      | R2 Cutoff Probability      | Sigma              |
|---------------------|----------------------------|----------------------------|--------------------|
| 2015                | 29.36%<br>(23.20%, 36.38%) | 39.20%<br>(32.75%, 46.05%) |                    |
| 2016                | 29.17%<br>(22.21%, 37.27%) | 40.53%<br>(33.55%, 47.92%) |                    |
| 2017                | 37.06%<br>(32.10%, 42.29%) | 41.69%<br>(34.83%, 48.90%) |                    |
| 2018                | 38.12%<br>(33.98%, 42.44%) | 39.30%<br>(31.80%, 47.33%) |                    |
| 2019                | 34.04%<br>(28.67%, 39.85%) | 39.62%<br>(32.54%, 47.16%) |                    |
| Sigma               |                            |                            | 0.0064<br>(0.0035) |
| Observations = 2329 | Parameters = 80            | LL = -2992.0               | AIC = 6144.1       |

## A3: Economics Department Static Estimates

### Table 9: Economics Department Static Estimates

(a) Unrestricted

(b) Restricted

[illegible][illegible]

## ►► Restricted Dynamic Estimates

### A3: Economics Department Myopic Estimates

Table 10: Economics Department Myopic Estimates

[illegible][illegible]

Robust standard errors in parentheses. <sup>\*</sup>p<0.10, <sup>\*\*</sup>p<0.05, <sup>\*\*\*</sup>p<0.01. Confidence intervals are 95%.

Robust standard errors in parentheses. <sup>\*</sup> $p < 0.10$ , <sup>\*\*</sup> $p < 0.05$ , <sup>\*\*\*</sup> $p < 0.01$ .

## A3: Economics Department Matriculation-Dynamic Estimates

Table 11: Economics Department Matriculation-Dynamic Estimates

(a) Unrestricted

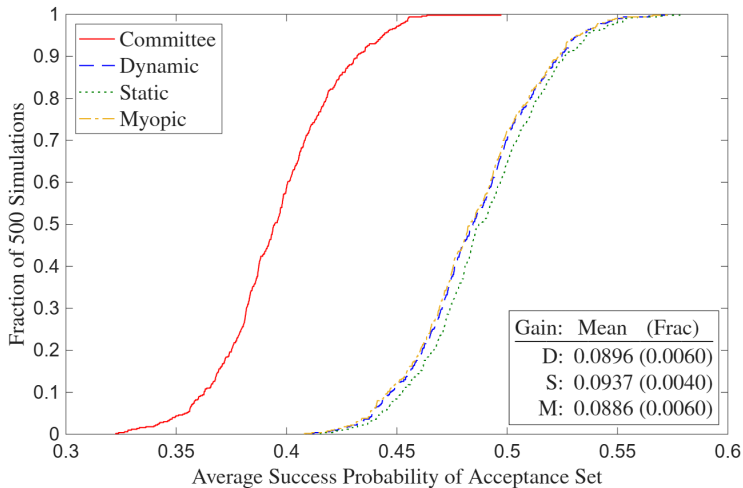
(b) Restricted

[illegible][illegible]

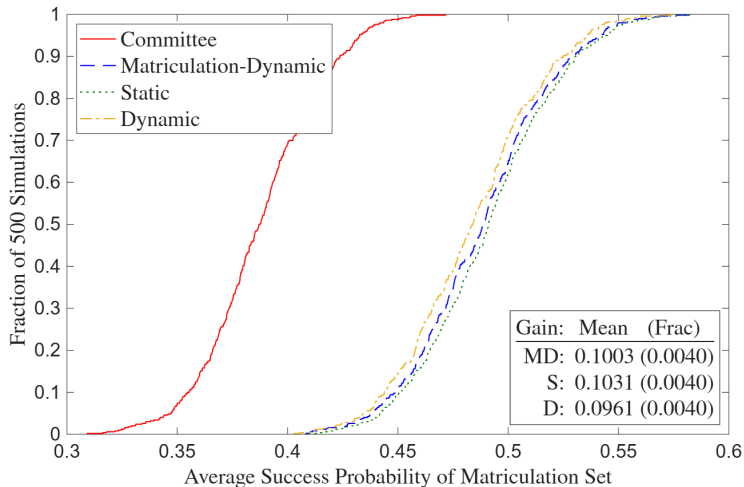
| City                                                                                                                                  | Param | IG | ΔIC | 2020 | 500 | 1,000 | 2003.0 | 2003.5 |
|---------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----|------|-----|-------|--------|--------|
| Robust standard errors in parentheses: <sup>*</sup> p<0.10, <sup>**</sup> p<0.05, <sup>***</sup> p<0.01. Confidence intervals are 95% |       |    |     |      |     |       |        |        |

## ► Restricted Matriculation-Dynamic Estimates

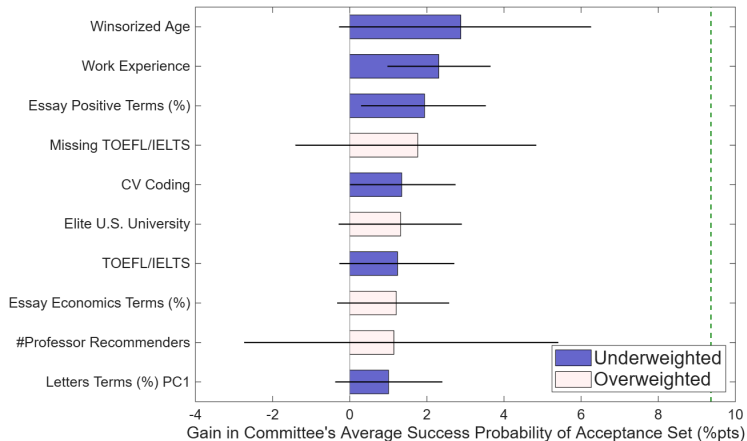
# A3: Economics Department Deviation Gains (Acceptance)



# A3: Economics Department Deviation Gains (Matriculation)

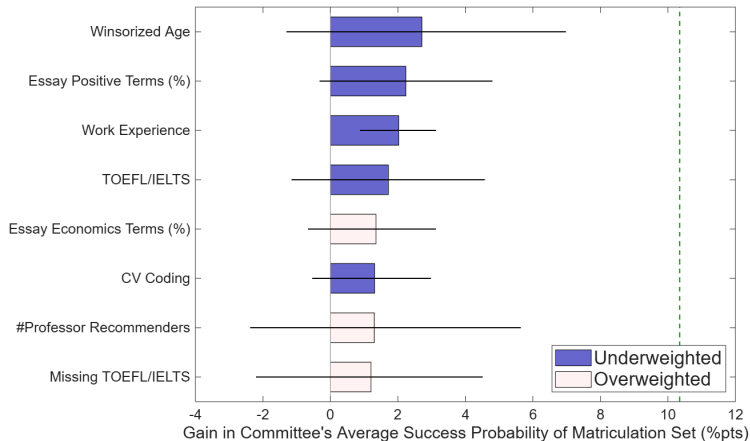


# A3: Economics Department Gains Attribution (Acceptance)



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

# A3: Economics Department Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

# A3: Economics Department Static Attribution

Table 12: Economics Department Static Attribution

|                           | Gain (%pt) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---------------------------|------------|-------------|----------------|-------------------------|--------------------|----------------------------|-----------------------|
| Winsorized Age            | 2.8788     | 25.9910     | 26.2840        | 29.9572                 | 29.5331            | 1.1674                     | 1.0326                |
| Work Experience           | 2.3065     | 0.5247      | 0.7276         | 0.9844                  | 0.9650             | 0.5141                     | 0.4752                |
| Essay Positive Terms (%)  | 1.9410     | 11.7499     | 11.6521        | 12.5376                 | 12.1183            | 0.7858                     | 0.4137                |
| Missing TOEFL/IELTS       | 1.7635     | 0.6076      | 0.6654         | 0.0739                  | 0.4086             | -1.2110                    | -0.5258               |
| CV Coding                 | 1.3478     | 2.8001      | 2.7820         | 3.3399                  | 2.9415             | 0.6244                     | 0.1785                |
| Elite U.S. University     | 1.3179     | 0.1267      | 0.2335         | 0.0078                  | 0.0389             | -0.6784                    | -0.5848               |
| TOEFL/IELTS               | 1.2400     | 7.4376      | 7.5675         | 7.7759                  | 7.6827             | 0.5957                     | 0.3294                |
| Essay Economics Terms (%) | 1.2043     | 9.7228      | 9.7423         | 9.0284                  | 9.5388             | -0.6411                    | -0.1828               |
| #Professor Recommenders   | 1.1455     | 2.6230      | 2.6148         | 1.3852                  | 2.1362             | -1.6680                    | -0.6493               |
| Letters Terms (%) PC1     | 1.0074     | -0.0106     | 0.1374         | 0.4969                  | 0.1387             | 0.5202                     | 0.0018                |

►► Deviation Gains Attribution

# A3: Economics Department Matriculation-Static Attribution

Table 13: Economics Department Matriculation-Static Attribution

|                           | Gain (%pt) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---------------------------|------------|-------------|----------------|-------------------------|--------------------|----------------------------|-----------------------|
| Winsorized Age            | 2.7099     | 25.9910     | 26.3284        | 30.9159                 | 29.7737            | 1.4580                     | 1.0950                |
| Essay Positive Terms (%)  | 2.2339     | 11.7499     | 11.6107        | 12.9136                 | 12.2763            | 1.1534                     | 0.5905                |
| Work Experience           | 2.0220     | 0.5247      | 0.7938         | 0.9949                  | 0.9799             | 0.4077                     | 0.3725                |
| TOEFL/IELTS               | 1.7179     | 7.4376      | 7.4939         | 7.8501                  | 7.6128             | 1.0238                     | 0.3397                |
| Essay Economics Terms (%) | 1.3503     | 9.7228      | 9.9994         | 9.0856                  | 9.5738             | -0.8271                    | -0.3821               |
| CV Coding                 | 1.3100     | 2.8001      | 2.6661         | 3.3395                  | 2.8782             | 0.7618                     | 0.2374                |
| #Professor Recommenders   | 1.3005     | 2.6230      | 2.6171         | 1.4173                  | 2.2255             | -1.6321                    | -0.5312               |
| Missing TOEFL/IELTS       | 1.1999     | 0.6076      | 0.6858         | 0.0442                  | 0.4368             | -1.3075                    | -0.5099               |

►► Deviation Gains Attribution (Matriculation)

# A3: Graduate School Means by Program

Table 14a: Graduate School Means by Program

|                                 | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Accept                          | 0.1121    | 0.0430     | 0.0986  | 0.0678      | 0.0535     |
| Covid Year                      | 0.2478    | 0.2792     | 0.3029  | 0.3460      | 0.3004     |
| Winsorized Age                  | 26.2782   | 27.9633    | 26.8478 | 28.0692     | 25.8230    |
| Female                          | 0.4201    | 0.4182     | 0.4232  | 0.6133      | 0.7490     |
| U.S. African/Hispanic/Native    | 0.0289    | 0.0766     | 0.1159  | 0.0665      | 0.1728     |
| U.S. Asian                      | 0.0313    | 0.0421     | 0.0304  | 0.0353      | 0.0576     |
| International Asian             | 0.5322    | 0.2510     | 0.1058  | 0.2714      | 0.1523     |
| International Other             | 0.2517    | 0.2967     | 0.2203  | 0.3256      | 0.1584     |
| GRE Verbal Percentile           | 71.8677   | 81.3486    | 82.6054 | 73.5451     | 75.6192    |
| GRE Quantitative Percentile     | 86.2163   | 63.3428    | 49.3295 | 57.8455     | 58.0000    |
| GRE Analytic Percentile         | 52.6351   | 72.1905    | 75.0460 | 63.6824     | 69.2077    |
| Undergraduate GPA               | 3.6534    | 3.5675     | 3.6531  | 3.6567      | 3.5695     |
| TOEFL/IELTS                     | 7.5232    | 7.5422     | 7.4252  | 7.5444      | 7.3158     |
| Elite U.S. University           | 0.1516    | 0.1847     | 0.2565  | 0.1954      | 0.2840     |
| Elite U.S. Liberal Arts College | 0.0423    | 0.0672     | 0.0696  | 0.0231      | 0.0535     |
| Elite Foreign University        | 0.0717    | 0.0578     | 0.0478  | 0.0488      | 0.0206     |
| Other Foreign University        | 0.5736    | 0.3953     | 0.2333  | 0.5102      | 0.1852     |
| Graduate Degree                 | 0.7478    | 0.7553     | 0.6652  | 0.7544      | 0.4259     |
| Work Experience                 | 0.5938    | 0.7463     | 0.6638  | 0.7544      | 0.7593     |
| #Professor Recommenders         | 2.5173    | 2.3469     | 2.6261  | 2.4355      | 2.1831     |
| #Prolific Recommenders          | 1.0760    | 0.6912     | 0.5043  | 0.7069      | 0.3128     |
| #Math Courses                   | 6.6396    |            |         |             |            |
| Advanced Math                   | 0.4827    |            |         |             |            |
| Missing GRE                     | 0.0255    | 0.4612     | 0.6217  | 0.6839      | 0.4650     |
| Missing Undergraduate GPA       | 0.6607    | 0.4827     | 0.3072  | 0.5726      | 0.2490     |
| Missing TOEFL/IELTS             | 0.5958    | 0.7714     | 0.8449  | 0.7707      | 0.8827     |
| Observations                    | 2078      | 2231       | 690     | 737         | 486        |

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

# A3: Graduate School Means by Program, Accept = 1

Table 14b: Graduate School Means by Program, Accept = 1

|                                 | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Covid Year                      | 0.1888    | 0.2188     | 0.2059  | 0.2200      | 0.2308     |
| Winsorized Age                  | 25.0815   | 27.7292    | 27.2794 | 27.1000     | 25.7692    |
| Female                          | 0.4850    | 0.5000     | 0.5588  | 0.5800      | 0.6154     |
| U.S. African/Hispanic/Native    | 0.0386    | 0.1458     | 0.1765  | 0.0400      | 0.0769     |
| U.S. Asian                      | 0.0429    | 0.0312     | 0.0147  | 0.0600      | 0.1154     |
| International Asian             | 0.4163    | 0.1667     | 0.0882  | 0.3600      | 0.0769     |
| International Other             | 0.2575    | 0.2292     | 0.3235  | 0.2000      | 0.1538     |
| GRE Verbal Percentile           | 84.3980   | 88.8725    | 86.6328 | 77.9161     | 85.7729    |
| GRE Quantitative Percentile     | 91.9289   | 68.3708    | 54.8113 | 64.5504     | 62.3462    |
| GRE Analytic Percentile         | 69.0355   | 82.7222    | 78.0879 | 68.8958     | 74.6169    |
| Undergraduate GPA               | 3.7336    | 3.6404     | 3.7204  | 3.7004      | 3.6341     |
| TOEFL/IELTS                     | 7.6419    | 7.6050     | 7.4730  | 7.6237      | 7.3178     |
| Elite U.S. University           | 0.2833    | 0.3125     | 0.3382  | 0.2200      | 0.5385     |
| Elite U.S. Liberal Arts College | 0.1030    | 0.1458     | 0.1176  | 0.0200      | 0.1154     |
| Elite Foreign University        | 0.1030    | 0.0521     | 0.0588  | 0.1000      | 0.0385     |
| Other Foreign University        | 0.3863    | 0.2396     | 0.2647  | 0.4000      | 0.1154     |
| Graduate Degree                 | 0.5837    | 0.6771     | 0.7206  | 0.7400      | 0.4615     |
| Work Experience                 | 0.6309    | 0.8125     | 0.7647  | 0.7000      | 0.9231     |
| #Professor Recommenders         | 2.5494    | 2.3750     | 2.8235  | 2.7200      | 2.2308     |
| #Prolific Recommenders          | 1.2833    | 0.9167     | 0.6765  | 1.0800      | 0.4615     |
| #Math Courses                   | 7.2575    |            |         |             |            |
| Advanced Math                   | 0.6781    |            |         |             |            |
| Missing GRE                     | 0.0086    | 0.2917     | 0.6324  | 0.5800      | 0.1923     |
| Missing Undergraduate GPA       | 0.4893    | 0.3125     | 0.3088  | 0.4800      | 0.1923     |
| Missing TOEFL/IELTS             | 0.6567    | 0.8854     | 0.8529  | 0.7600      | 0.8846     |
| Observations                    | 233       | 96         | 68      | 50          | 26         |

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

# A3: Graduate School Correlation Matrix

Table 15: Graduate School Correlation Matrix

|                                 | Covid | Age   | Fem   | USAHN | USAsn | IntAsn | IntOth | GREV  | GREQ  | GREa  | GPA   | T/I   | EUSUni | EUSLA | EFor  | OthFor | Grad  | Work  | Prof  | Pro   | MGRE  | MGPA  | MT/I |
|---------------------------------|-------|-------|-------|-------|-------|--------|--------|-------|-------|-------|-------|-------|--------|-------|-------|--------|-------|-------|-------|-------|-------|-------|------|
| Covid Year                      | 1.00  |       |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Winsorized Age                  | 0.01  | 1.00  |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Female                          | 0.02  | -0.13 | 1.00  |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| U.S. African/Hispanic/Native    | 0.00  | -0.04 | 0.04  | 1.00  |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| U.S. Asian                      | -0.00 | -0.06 | 0.05  | -0.05 | 1.00  |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| International Asian             | -0.05 | -0.07 | 0.02  | -0.19 | -0.14 | 1.00   |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| International Other             | 0.01  | 0.18  | -0.06 | -0.17 | -0.12 | -0.42  | 1.00   |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Verbal Percentile           | 0.03  | -0.06 | -0.04 | 0.00  | 0.04  | -0.12  | -0.15  | 1.00  |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Quantitative Percentile     | -0.02 | -0.16 | -0.10 | -0.18 | 0.00  | 0.44   | -0.10  | 0.16  | 1.00  |       |       |       |        |       |       |        |       |       |       |       |       |       |      |
| GRE Analytic Percentile         | 0.05  | -0.06 | 0.03  | 0.08  | 0.08  | -0.37  | -0.08  | 0.60  | -0.20 | 1.00  |       |       |        |       |       |        |       |       |       |       |       |       |      |
| Undergraduate GPA               | -0.02 | -0.29 | 0.07  | -0.13 | -0.02 | 0.02   | 0.00   | 0.06  | 0.12  | 0.05  | 1.00  |       |        |       |       |        |       |       |       |       |       |       |      |
| TOEFL/IELTS                     | -0.01 | -0.07 | 0.01  | -0.02 | 0.01  | 0.04   | 0.00   | 0.24  | 0.12  | 0.19  | 0.00  | 1.00  |        |       |       |        |       |       |       |       |       |       |      |
| Elite U.S. University           | 0.01  | -0.12 | 0.02  | 0.12  | 0.14  | -0.12  | -0.22  | 0.16  | -0.00 | 0.19  | -0.01 | -0.02 | 1.00   |       |       |        |       |       |       |       |       |       |      |
| Elite U.S. Liberal Arts College | -0.02 | -0.02 | 0.04  | 0.04  | 0.04  | -0.08  | -0.09  | 0.12  | -0.01 | 0.13  | -0.02 | -0.00 | -0.11  | 1.00  |       |        |       |       |       |       |       |       |      |
| Elite Foreign University        | -0.00 | -0.04 | -0.00 | -0.06 | -0.04 | 0.09   | 0.09   | 0.02  | 0.08  | -0.02 | 0.01  | 0.04  | -0.12  | -0.06 | 1.00  |        |       |       |       |       |       |       |      |
| Other Foreign University        | -0.02 | 0.20  | -0.04 | -0.23 | -0.14 | 0.34   | 0.39   | -0.26 | 0.20  | -0.39 | 0.02  | 0.02  | -0.43  | -0.21 | -0.22 | 1.00   |       |       |       |       |       |       |      |
| Graduate Degree                 | 0.00  | 0.41  | -0.08 | -0.13 | -0.09 | 0.21   | 0.15   | -0.14 | 0.05  | -0.20 | -0.22 | 0.03  | -0.21  | -0.10 | 0.05  | 0.37   | 1.00  |       |       |       |       |       |      |
| Work Experience                 | 0.01  | 0.33  | 0.05  | 0.05  | 0.01  | -0.18  | 0.11   | 0.01  | -0.15 | 0.11  | -0.08 | -0.03 | -0.02  | 0.03  | -0.03 | -0.01  | 0.10  | 1.00  |       |       |       |       |      |
| #Professor Recommenders         | -0.02 | -0.19 | -0.04 | -0.04 | 0.01  | 0.11   | -0.09  | 0.01  | 0.09  | -0.04 | 0.14  | 0.03  | -0.02  | 0.02  | -0.00 | -0.02  | -0.03 | -0.18 | 1.00  |       |       |       |      |
| #Prolific Recommenders          | 0.01  | -0.06 | -0.02 | -0.07 | -0.01 | 0.24   | -0.10  | -0.01 | 0.25  | -0.11 | 0.04  | 0.10  | -0.01  | -0.01 | 0.06  | 0.08   | 0.17  | -0.09 | 0.19  | 1.00  |       |       |      |
| Missing GRE                     | 0.09  | 0.15  | 0.05  | 0.11  | -0.01 | -0.22  | 0.17   | 0.09  | -0.33 | 0.17  | -0.09 | -0.10 | -0.03  | -0.04 | -0.03 | -0.01  | 0.02  | 0.10  | -0.08 | -0.20 | 1.00  |       |      |
| Missing Undergraduate GPA       | -0.01 | 0.18  | -0.04 | -0.24 | -0.15 | 0.36   | 0.42   | -0.24 | 0.21  | -0.38 | 0.02  | 0.04  | -0.46  | -0.23 | 0.23  | 0.83   | 0.37  | -0.02 | -0.03 | 0.10  | -0.01 | 1.00  |      |
| Missing TOEFL/IELTS             | 0.05  | -0.07 | 0.03  | 0.16  | 0.11  | -0.32  | -0.22  | 0.16  | -0.24 | 0.32  | -0.02 | -0.02 | 0.29   | 0.14  | -0.04 | -0.58  | -0.20 | 0.05  | -0.07 | -0.05 | 0.12  | -0.58 | 1.00 |
| Observations                    | 6222  |       |       |       |       |        |        |       |       |       |       |       |        |       |       |        |       |       |       |       |       |       |      |

# A3: Graduate School Logistic Regression Results

Table 16a: Graduate School Logistic Regression Results

| Dependent Variable = Accept  | Economics              | Government            | History               | Linguistics           | Psychology            |
|------------------------------|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Covid Year                   | -0.3966**<br>(0.2010)  | -0.4332*<br>(0.2612)  | -0.5792**<br>(0.3470) | -0.5501<br>(0.4120)   | -0.1766<br>(0.5545)   |
| Winsorized Age               | -0.0995***<br>(0.0358) | 0.0454*<br>(0.0271)   | 0.0380<br>(0.0327)    | 0.0222<br>(0.0532)    | -0.0520<br>(0.0890)   |
| Female                       | 0.4079**<br>(0.1681)   | 0.6721***<br>(0.2299) | 0.6539**<br>(0.2971)  | 0.1311<br>(0.3452)    | 0.5660<br>(0.5606)    |
| U.S. African/Hispanic/Native | 0.4221<br>(0.5258)     | 1.0081***<br>(0.3545) | 1.1601***<br>(0.4426) | 0.0120<br>(0.8411)    | -0.8263<br>(1.1511)   |
| U.S. Asian                   | -0.5394<br>(0.4190)    | -0.5660<br>(0.6504)   | -0.3250<br>(1.2512)   | 0.8638<br>(0.7201)    | 1.1711<br>(0.7276)    |
| International Asian          | -0.8445***<br>(0.2945) | -0.1853<br>(0.4146)   | 0.9405<br>(0.7233)    | 0.4890<br>(0.6720)    | -0.7827<br>(0.8323)   |
| International Other          | 0.5034<br>(0.3172)     | 0.3781<br>(0.4131)    | 1.3529***<br>(0.5086) | -0.3061<br>(0.7483)   | 0.6239<br>(1.1070)    |
| GRE Verbal Percentile        | 0.0155***<br>(0.0050)  | 0.0281*<br>(0.0160)   | 0.0753**<br>(0.0344)  | -0.0092<br>(0.0154)   | 0.9607***<br>(0.0218) |
| GRE Quantitative Percentile  | 0.1547***<br>(0.0186)  | 0.0161*<br>(0.0086)   | 0.0219*<br>(0.0116)   | 0.0211<br>(0.0150)    | -0.0138<br>(0.0159)   |
| GRE Analytic Percentile      | 0.0179***<br>(0.0042)  | 0.0199**<br>(0.0094)  | 0.0030<br>(0.0134)    | 0.0175<br>(0.0127)    | -0.0202<br>(0.0218)   |
| Undergraduate GPA            | 2.2212***<br>(0.7219)  | 0.6969<br>(0.5476)    | 1.3849**<br>(0.5513)  | 1.2646<br>(0.9703)    | 1.7722*<br>(0.9487)   |
| TOEFL/IELTS                  | 0.3878<br>(0.2883)     | 0.6456<br>(0.4708)    | 1.0963**<br>(0.5526)  | 0.9655*<br>(0.5496)   | -0.4656<br>(0.6112)   |
| Elite U.S. University        | 0.8850***<br>(0.2960)  | 0.3943<br>(0.3135)    | 0.5541*<br>(0.3802)   | 0.6657<br>(0.4085)    | 1.4303**<br>(0.6102)  |
| Elite U.S. Liberal Arts      | 1.1173***<br>(0.3813)  | 0.6279*<br>(0.3754)   | 1.0698**<br>(0.5190)  | -0.6088<br>(1.4106)   | 2.3151**<br>(1.0842)  |
| Elite Foreign University     | 1.0061*<br>(0.6050)    | 0.4759*<br>(0.2224)   | 0.7595*<br>(0.7938)   | 1.7718***<br>(0.8587) | 1.6165*<br>(1.9880)   |
| Other Foreign University     | 0.5442<br>(0.5785)     | 0.5777<br>(0.7688)    | 1.3733**<br>(0.6902)  | 0.4188<br>(0.7157)    | -1.0737<br>(1.3016)   |
| Graduate Degree              | 0.0467<br>(0.2238)     | 0.0733<br>(0.2583)    | 0.3128<br>(0.3280)    | 0.4849<br>(0.4896)    | 0.4892<br>(0.6390)    |
| Work Experience              | 0.3932**<br>(0.1889)   | 0.4027<br>(0.2993)    | 0.4972<br>(0.3295)    | 0.1495<br>(0.3778)    | 1.8201**<br>(0.8564)  |
| #Professor Recommenders      | 0.0346<br>(0.1317)     | -0.0178<br>(0.1339)   | 0.5511***<br>(0.2554) | -0.4701*<br>(0.2652)  | -0.0981<br>(0.2660)   |
| #Prolific Recommenders       | 0.2635***<br>(0.0854)  | 0.3025**<br>(0.1262)  | 0.1880<br>(0.1576)    | 0.3284**<br>(0.1544)  | 0.5001*<br>(0.3145)   |
| #Math Courses                | 0.1689***<br>(0.0596)  |                       |                       |                       |                       |
| Advanced Math                | 0.2319<br>(0.1790)     |                       |                       |                       |                       |
| Missing GRE                  | 0.1899<br>(0.7646)     | -0.3259<br>(0.3100)   | 0.6282<br>(0.4275)    | -0.0831<br>(0.3761)   | -1.1514**<br>(0.5702) |
| Missing Undergraduate GPA    | -0.4745<br>(0.4891)    | -0.4017<br>(0.5440)   | -1.5492**<br>(0.6893) | -1.1087*<br>(0.6401)  | 0.4890<br>(1.1831)    |
| Missing TOEFL/IELTS          | -0.3786<br>(0.2432)    | 0.6188<br>(0.5657)    | 0.6745<br>(0.5190)    | -0.0440<br>(0.5099)   | -1.7224**<br>(0.9271) |
| Concentration FE             | No                     | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                 | 2078                   | 2231                  | 690                   | 737                   | 486                   |
| Pseudo-R <sup>2</sup>        | 0.2677                 | 0.1525                | 0.1962                | 0.1871                | 0.2475                |

Robust standard errors in parentheses. \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive** ( $p < .1$ ): Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative** ( $p < .1$ ): Covid Year, Winsorized Age, International Asian

# A3: Graduate School Logistic Regression AMEs

Table 16b: Graduate School Logistic Regression AMEs

| Dependent Variable = Accept     | Economics              | Government            | History               | Linguistics           | Psychology             | Non-Economics          |
|---------------------------------|------------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|
| Covid Year                      | -0.0303**<br>(0.0153)  | -0.0164<br>(0.0100)   | -0.0431**<br>(0.0261) | -0.0298<br>(0.0224)   | -0.0075<br>(0.0238)    | -0.0218***<br>(0.0084) |
| Winsorized Age                  | 0.0017**<br>(0.0027)   | 0.0017**<br>(0.0010)  | 0.0028<br>(0.0025)    | 0.0012<br>(0.0020)    | -0.0022<br>(0.0038)    | 0.0016*<br>(0.0009)    |
| Female                          | 0.0312***<br>(0.0129)  | 0.0254***<br>(0.0088) | 0.0400***<br>(0.0220) | 0.0071<br>(0.0187)    | -0.0216<br>(0.0239)    | 0.0201***<br>(0.0075)  |
| U.S. African/Hispanic/Native    | 0.0323<br>(0.0402)     | 0.0381***<br>(0.0137) | 0.0869***<br>(0.0332) | 0.0007<br>(0.0455)    | -0.0352<br>(0.0487)    | 0.0384***<br>(0.0119)  |
| U.S. Asian                      | -0.0413<br>(0.0322)    | -0.0215<br>(0.0247)   | -0.0204<br>(0.0936)   | 0.0468<br>(0.0391)    | 0.0499<br>(0.0513)     | 0.0052<br>(0.0185)     |
| International Asian             | -0.0646***<br>(0.0224) | -0.0070<br>(0.0157)   | 0.0704<br>(0.0543)    | 0.0265<br>(0.0363)    | -0.0334<br>(0.0393)    | 0.0069<br>(0.0132)     |
| International Other             | 0.0385<br>(0.0423)     | 0.0143<br>(0.0157)    | 0.1013***<br>(0.0385) | -0.0057<br>(0.0406)   | 0.0266<br>(0.0471)     | 0.0238*<br>(0.0132)    |
| GRE Verbal Percentile           | 0.0012***<br>(0.0005)  | 0.0011*<br>(0.0006)   | 0.0056**<br>(0.0025)  | -0.0005<br>(0.0008)   | 0.0026***<br>(0.0010)  | 0.0015***<br>(0.0005)  |
| GRE Quantitative Percentile     | 0.0118***<br>(0.0014)  | 0.0056*<br>(0.0003)   | 0.0016*<br>(0.0009)   | 0.0011<br>(0.0008)    | -0.0006<br>(0.0007)    | 0.0005**<br>(0.0003)   |
| GRE Analytic Percentile         | 0.0014***<br>(0.0003)  | 0.0008**<br>(0.0004)  | 0.0002<br>(0.0010)    | -0.0009<br>(0.0007)   | -0.0009<br>(0.0009)    | 0.0007**<br>(0.0003)   |
| Undergraduate GPA               | 0.1699***<br>(0.0551)  | 0.0264<br>(0.0308)    | 0.1022**<br>(0.0423)  | 0.0984<br>(0.0522)    | 0.0757*<br>(0.0418)    | 0.0485***<br>(0.0176)  |
| TOEFL/IELTS                     | 0.0207<br>(0.0219)     | 0.0044<br>(0.0179)    | 0.0523*<br>(0.0417)   | 0.0529*<br>(0.0301)   | -0.0198<br>(0.0259)    | 0.0326***<br>(0.0123)  |
| Elite U.S. University           | 0.0277***<br>(0.0225)  | 0.0149<br>(0.0119)    | 0.0610**<br>(0.0281)  | 0.0461***<br>(0.0264) | -0.0233***<br>(0.0271) | 0.0409**<br>(0.0098)   |
| Elite U.S. Liberal Arts College | 0.0859***<br>(0.0288)  | 0.0238*<br>(0.0143)   | 0.0801**<br>(0.0402)  | -0.0130<br>(0.0761)   | 0.0667**<br>(0.0480)   | 0.0420***<br>(0.0136)  |
| Elite Foreign University        | 0.0180<br>(0.0463)     | 0.0770**<br>(0.0311)  | 0.0677*<br>(0.0591)   | 0.0166<br>(0.0481)    | 0.0479***<br>(0.0650)  | 0.0063<br>(0.0241)     |
| Other Foreign University        | 0.0416<br>(0.0441)     | 0.0219<br>(0.0291)    | 0.1028*<br>(0.0527)   | 0.0227<br>(0.0391)    | -0.0458<br>(0.0556)    | 0.0273<br>(0.0221)     |
| Graduate Degree                 | 0.0026<br>(0.0171)     | 0.0028<br>(0.0098)    | 0.0234<br>(0.0247)    | 0.0219<br>(0.0262)    | 0.0208<br>(0.0273)     | 0.0106<br>(0.0089)     |
| Work Experience                 | 0.0301**<br>(0.0144)   | 0.0152<br>(0.0113)    | 0.0372<br>(0.0248)    | 0.0081<br>(0.0205)    | 0.0776**<br>(0.0368)   | 0.0221**<br>(0.0089)   |
| #Professor Recommenders         | 0.0005<br>(0.0101)     | -0.0007<br>(0.0051)   | 0.0209*<br>(0.0190)   | 0.0209*<br>(0.0142)   | -0.0042<br>(0.0113)    | 0.0063<br>(0.0047)     |
| #Prolific Recommenders          | 0.0202***<br>(0.0065)  | 0.0114**<br>(0.0048)  | 0.0141<br>(0.0118)    | 0.0178**<br>(0.0082)  | 0.0251*<br>(0.0130)    | 0.0133***<br>(0.0038)  |
| #Math Courses                   | 0.0129***<br>(0.0045)  |                       |                       |                       |                        |                        |
| Advanced Math                   | 0.0177<br>(0.0137)     |                       |                       |                       |                        |                        |
| Missing GRE                     | 0.0145<br>(0.0585)     | -0.0123<br>(0.0118)   | 0.0470<br>(0.0319)    | -0.0045<br>(0.0204)   | -0.0401*<br>(0.0252)   | -0.0062<br>(0.0088)    |
| Missing Undergraduate GPA       | -0.0363<br>(0.0375)    | -0.0186<br>(0.0266)   | -0.1160**<br>(0.0524) | -0.0601*<br>(0.0355)  | 0.0200<br>(0.0605)     | -0.0291<br>(0.0179)    |
| Missing TOEFL/IELTS             | -0.0290<br>(0.0188)    | 0.0234<br>(0.0215)    | 0.0505<br>(0.0388)    | -0.0034<br>(0.0276)   | -0.0734*<br>(0.0417)   | 0.0108<br>(0.0135)     |
| Concentration FE                | No                     | Yes                   | Yes                   | Yes                   | Yes                    | Yes                    |
| Observations                    | 2078                   | 2231                  | 690                   | 737                   | 486                    | 4144                   |
| Wald Statistic   DF   P-Value   | 115.8637               | 23                    | 0.0000                |                       |                        |                        |

Robust standard errors in parentheses. \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- Economics Positive ( $p < .1$ ):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- Economics Negative ( $p < .1$ ):** Covid Year, Winsorized Age, International Asian

# A3: Graduate School Logistic Regression Results (Synthetic)

Table 17a: Graduate School Logistic Regression Results (Synthetic)

| Dependent Variable = Accept     | Economics             | Government           | History               | Linguistics            | Psychology             |
|---------------------------------|-----------------------|----------------------|-----------------------|------------------------|------------------------|
| Covid Year                      | -0.1997<br>(0.1633)   | -0.0223<br>(0.2215)  | 0.2028<br>(0.3078)    | -0.4791<br>(0.3477)    | -0.5946<br>(0.4520)    |
| Winsorized Age                  | -0.0509**<br>(0.0201) | -0.0198<br>(0.0174)  | 0.0576*<br>(0.0307)   | -0.0164<br>(0.0357)    | 0.1203***<br>(0.0400)  |
| Female                          | 0.1941<br>(0.1402)    | 0.0923<br>(0.2004)   | 0.1389<br>(0.2526)    | 0.2595<br>(0.3255)     | 0.2119<br>(0.4310)     |
| U.S. African/Hispanic/Native    | 0.3053<br>(0.3724)    | -0.1457<br>(0.4305)  | -0.2379<br>(0.4670)   | -1.6401<br>(1.0720)    | 0.5413<br>(0.5306)     |
| U.S. Asian                      | 0.5663<br>(0.3642)    | 0.6016<br>(0.4452)   | -0.7804<br>(1.1675)   | -0.4305<br>(0.7036)    | 0.7122<br>(0.8863)     |
| International Asian             | -0.1428<br>(0.2169)   | 0.1630<br>(0.2803)   | 0.5780<br>(0.3740)    | -0.4065<br>(0.4275)    | 0.1868<br>(0.6866)     |
| International Other             | -0.1231<br>(0.2111)   | 0.1395<br>(0.3046)   | 0.1097<br>(0.3237)    | -0.1602<br>(0.3832)    | -1.0807<br>(0.7583)    |
| GRE Verbal Percentile           | 0.0167***<br>(0.0040) | 0.0247**<br>(0.0117) | 0.0164<br>(0.0127)    | -0.0029<br>(0.0100)    | 0.0132<br>(0.0184)     |
| GRE Quantitative Percentile     | 0.0233***<br>(0.0090) | -0.0027<br>(0.0051)  | 0.0026<br>(0.0088)    | 0.0304**<br>(0.0137)   | 0.0097<br>(0.0090)     |
| GRE Analytic Percentile         | 0.0042<br>(0.0031)    | 0.0004<br>(0.0050)   | -0.0073<br>(0.0092)   | -0.0042<br>(0.0101)    | 0.0069<br>(0.0103)     |
| Undergraduate GPA               | 0.2292<br>(0.4745)    | 0.6327<br>(0.4085)   | 1.3552***<br>(0.5796) | -1.3171**<br>(0.5584)  | -0.1002<br>(0.6152)    |
| TOEFL/IELTS                     | -0.4673**<br>(0.2157) | 0.5141<br>(0.4185)   | -0.7648<br>(0.6814)   | 0.2446<br>(0.7498)     | -0.7354<br>(0.8025)    |
| Elite U.S. University           | -0.4850*<br>(0.2560)  | -0.2243<br>(0.3258)  | 0.0572<br>(0.3421)    | 0.4764<br>(0.4788)     | 0.3941<br>(0.4782)     |
| Elite U.S. Liberal Arts College | 0.2429<br>(0.3049)    | 0.1750<br>(0.4042)   | 0.2358<br>(0.4964)    | 0.4860<br>(0.9631)     | 0.9652<br>(0.7261)     |
| Elite Foreign University        | -0.1087<br>(0.3015)   | -0.1622<br>(0.7484)  | -0.1029<br>(0.5668)   | 0.4801<br>(0.7841)     | 3.0516***<br>(0.8673)  |
| Other Foreign University        | -0.1087<br>(0.1979)   | 0.2043<br>(0.2732)   | 0.2721<br>(0.3391)    | 0.1731<br>(0.4202)     | 0.3023<br>(0.5926)     |
| Graduate Degree                 | -0.0186<br>(0.1624)   | -0.0186<br>(0.2439)  | -0.1661<br>(0.2765)   | -0.0698***<br>(0.3320) | -0.2769<br>(0.3814)    |
| Work Experience                 | 0.2525*<br>(0.1441)   | -0.0391<br>(0.2396)  | -0.0736<br>(0.2629)   | -0.2650<br>(0.3417)    | -1.1936**<br>(0.4605)  |
| #Professor Recommenders         | -0.1059<br>(0.0909)   | 0.0911<br>(0.1138)   | -0.1838<br>(0.1473)   | 0.2161<br>(0.1817)     | -0.4924**<br>(0.1956)  |
| #Profic Recommenders            | 0.0039<br>(0.0636)    | 0.2189**<br>(0.1108) | -0.1743<br>(0.1723)   | 0.0673<br>(0.1642)     | 0.5134**<br>(0.2213)   |
| #Math Courses                   | 0.0396<br>(0.0421)    |                      |                       |                        |                        |
| Advanced Math                   | 0.1961<br>(0.1416)    |                      |                       |                        |                        |
| Missing GRE                     | -0.1093<br>(0.4328)   | -0.2029<br>(0.2195)  | -0.0081<br>(0.2689)   | -0.0433<br>(0.3503)    | -0.6040<br>(0.4017)    |
| Missing Undergraduate GPA       | 0.1531<br>(0.1540)    | 0.0895<br>(0.2293)   | -0.2455<br>(0.3015)   | 0.6298**<br>(0.3125)   | -0.3574<br>(0.4342)    |
| Missing TOEFL/IELTS             | -0.2909*<br>(0.1640)  | 0.4698<br>(0.3480)   | 0.5752<br>(0.5466)    | 0.1346<br>(0.4345)     | -0.0781***<br>(0.7633) |
| Concentration FE                | No                    | Yes                  | Yes                   | Yes                    | Yes                    |
| Observations                    | 2078                  | 2251                 | 690                   | 737                    | 486                    |
| Pseudo-R <sup>2</sup>           | 0.0524                | 0.0366               | 0.0753                | 0.1046                 | 0.2230                 |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive** ( $p < .1$ ): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ( $p < .1$ ): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

# A3: Graduate School Logistic Regression AMEs (Synthetic)

Table 17b: Graduate School Logistic Regression AMEs (Synthetic)

| Dependent Variable = Accept     | Economics             | Government           | History               | Linguistics            | Psychology             | Non-Economics        |
|---------------------------------|-----------------------|----------------------|-----------------------|------------------------|------------------------|----------------------|
| Covid Year                      | -0.0204<br>(0.0167)   | -0.0014<br>(0.0099)  | 0.0192<br>(0.0296)    | -0.0298<br>(0.0216)    | -0.0345<br>(0.0262)    | -0.0089<br>(0.0086)  |
| Winsorized Age                  | -0.0052**<br>(0.0021) | -0.0009<br>(0.0038)  | 0.0054**<br>(0.0029)  | -0.0010<br>(0.0022)    | 0.0070***<br>(0.0023)  | 0.0004<br>(0.0006)   |
| Female                          | 0.0198<br>(0.0143)    | 0.0041<br>(0.0089)   | 0.0162<br>(0.0239)    | 0.0123<br>(0.0204)     | 0.0099<br>(0.0251)     | 0.0099<br>(0.0078)   |
| U.S. African/Hispanic/Native    | 0.0312<br>(0.0381)    | -0.0065<br>(0.0192)  | -0.0225<br>(0.0441)   | -0.1021<br>(0.0673)    | 0.0314<br>(0.0396)     | -0.0137<br>(0.0147)  |
| U.S. Asian                      | 0.0679<br>(0.0372)    | 0.0266<br>(0.0199)   | -0.0750<br>(0.1105)   | -0.0268<br>(0.0495)    | 0.0413<br>(0.0516)     | 0.0241<br>(0.0179)   |
| International Asian             | -0.0146<br>(0.0222)   | 0.0073<br>(0.0125)   | 0.0546<br>(0.0353)    | -0.0309<br>(0.0263)    | 0.0108<br>(0.0399)     | 0.0075<br>(0.0112)   |
| International Other             | -0.0126<br>(0.0236)   | 0.0062<br>(0.0136)   | 0.0104<br>(0.0306)    | -0.0100<br>(0.0237)    | -0.0027<br>(0.0443)    | -0.0014<br>(0.0106)  |
| GRE Verbal Percentile           | 0.0017***<br>(0.0005) | 0.0011**<br>(0.0005) | 0.0015<br>(0.0012)    | -0.0002<br>(0.0006)    | 0.0008<br>(0.0010)     | 0.0008**<br>(0.0004) |
| GRE Quantitative Percentile     | 0.0024***<br>(0.0009) | -0.0001<br>(0.0002)  | 0.0003<br>(0.0008)    | 0.0019***<br>(0.0009)  | 0.0006<br>(0.0006)     | 0.0003<br>(0.0002)   |
| GRE Analytic Percentile         | 0.0004<br>(0.0003)    | 0.0000<br>(0.0003)   | -0.0007<br>(0.0009)   | -0.0003<br>(0.0006)    | 0.0004<br>(0.0006)     | 0.0000<br>(0.0002)   |
| Undergraduate GPA               | 0.0234<br>(0.0485)    | 0.0282<br>(0.0223)   | 0.1281***<br>(0.0553) | 0.0820**<br>(0.0398)   | 0.0016<br>(0.0357)     | 0.0130<br>(0.0158)   |
| TOEFL/IELTS                     | -0.0478**<br>(0.0221) | 0.0229<br>(0.0187)   | -0.0723<br>(0.0645)   | 0.0152<br>(0.0466)     | -0.0427<br>(0.0469)    | 0.0050<br>(0.0187)   |
| Elite U.S. University           | -0.0466*<br>(0.0261)  | -0.0100<br>(0.0145)  | 0.0054<br>(0.0324)    | 0.0297<br>(0.0297)     | 0.0229<br>(0.0274)     | 0.0036<br>(0.0108)   |
| Elite U.S. Liberal Arts College | 0.0248<br>(0.0312)    | 0.0078<br>(0.0180)   | 0.0223<br>(0.0469)    | 0.0303<br>(0.0602)     | 0.0502<br>(0.0410)     | 0.0217<br>(0.0160)   |
| Elite Foreign University        | -0.0166<br>(0.0308)   | -0.0512<br>(0.0337)  | 0.0700<br>(0.0537)    | 0.0299<br>(0.0400)     | 0.1771***<br>(0.0484)  | 0.0156<br>(0.0179)   |
| Other Foreign University        | -0.0111<br>(0.0202)   | 0.0091<br>(0.0122)   | 0.0257<br>(0.0321)    | 0.0108<br>(0.0267)     | 0.0175<br>(0.0344)     | 0.0116<br>(0.0103)   |
| Graduate Degree                 | -0.0020<br>(0.0166)   | 0.0079<br>(0.0109)   | -0.0015<br>(0.0261)   | -0.0642***<br>(0.0206) | -0.0161<br>(0.0220)    | -0.0063<br>(0.0085)  |
| Work Experience                 | 0.0258*<br>(0.0146)   | -0.0017<br>(0.0107)  | -0.0070<br>(0.0248)   | -0.0165<br>(0.0213)    | -0.0692**<br>(0.0275)  | -0.0093<br>(0.0086)  |
| #Professor Recommenders         | -0.0108<br>(0.0093)   | 0.0041<br>(0.0051)   | -0.0173<br>(0.0140)   | 0.0136<br>(0.0114)     | -0.0265***<br>(0.0110) | -0.0027<br>(0.0042)  |
| #Profic Recommenders            | 0.0004<br>(0.0097)    | 0.0096*<br>(0.0050)  | -0.0165<br>(0.0164)   | 0.0042<br>(0.0102)     | 0.0298**<br>(0.0126)   | 0.0033*<br>(0.0045)  |
| #Math Courses                   | 0.0037<br>(0.0043)    | 0.0037<br>(0.0043)   |                       |                        |                        |                      |
| Advanced Math                   | 0.0201<br>(0.0145)    |                      |                       |                        |                        |                      |
| Missing GRE                     | -0.0204<br>(0.0443)   | -0.0090<br>(0.0098)  | -0.0001<br>(0.0254)   | -0.0027<br>(0.0218)    | -0.0350<br>(0.0239)    | -0.0008<br>(0.0079)  |
| Missing Undergraduate GPA       | 0.0157<br>(0.0158)    | 0.0040<br>(0.0102)   | -0.0232<br>(0.0285)   | 0.0392*<br>(0.0201)    | -0.0207<br>(0.0254)    | 0.0019<br>(0.0063)   |
| Missing TOEFL/IELTS             | -0.0305*<br>(0.0168)  | 0.0218<br>(0.0156)   | 0.0544<br>(0.0517)    | 0.0684<br>(0.0272)     | -0.1265***<br>(0.0434) | 0.0063<br>(0.0130)   |
| Concentration FE                | No                    | Yes                  | Yes                   | Yes                    | Yes                    | Yes                  |
| Observations                    | 2078                  | 2231                 | 690                   | 737                    | 486                    | 4144                 |
| Wald Statistic   DF   P-Value   | 37.4658               | 23                   | 0.0290                |                        |                        |                      |

Robust standard errors in parentheses \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

- **Economics Positive ( $p < .1$ ):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ( $p < .1$ ):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

# A3: Graduate School Lasso Logistic Regression Results

Table 18a: Graduate School Lasso Logistic Regression Results

| Dependent Variable = Accept     | Economics | Government | History | Linguistics | Psychology |
|---------------------------------|-----------|------------|---------|-------------|------------|
| Covid Year                      | -0.3330   | -0.3123    | -0.3498 | -0.2378     |            |
| Winsorized Age                  | -0.0788   | 0.0266     | 0.0200  |             |            |
| Female                          | 0.3352    | 0.5096     | 0.5068  |             | -0.0654    |
| U.S. African/Hispanic/Native    | 0.2638    | 0.8041     | 0.7157  |             |            |
| U.S. Asian                      | -0.2802   | -0.2918    |         | 0.0413      | 0.1155     |
| International Asian             | -0.6744   | -0.0726    |         | 0.0465      |            |
| International Other             | 0.4729    | 0.1946     | 0.7240  | -0.1087     |            |
| GRE Verbal Percentile           | 0.0139    | 0.0252     | 0.0272  |             | 0.0343     |
| GRE Quantitative Percentile     | 0.1349    | 0.0119     | 0.0187  | 0.0154      |            |
| GRE Analytic Percentile         | 0.0166    | 0.0208     |         | 0.0080      |            |
| Undergraduate GPA               | 1.9977    | 0.4621     | 0.7414  | 0.1714      |            |
| TOEFL/IELTS                     | 0.4009    | 0.3295     | 0.5199  | 0.5151      |            |
| Elite U.S. University           | 0.6886    | 0.2709     | 0.3267  |             | 0.4606     |
| Elite U.S. Liberal Arts College | 0.8956    | 0.5199     | 0.5985  |             | 0.4470     |
| Elite Foreign University        | 0.3724    |            | 0.0414  | 0.4343      |            |
| Other Foreign University        |           |            |         |             |            |
| Graduate Degree                 |           |            | 0.0978  |             |            |
| Work Experience                 | 0.2835    | 0.3447     | 0.3273  |             | 0.5559     |
| #Professor Recommenders         |           |            | 0.3739  | 0.2322      |            |
| #Prolific Recommenders          | 0.2243    | 0.2362     | 0.1651  | 0.2213      | 0.0632     |
| #Math Courses                   | 0.1493    |            |         |             |            |
| Advanced Math                   | 0.2272    |            |         |             |            |
| Missing GRE                     |           | -0.2304    |         | -0.0527     | -0.5416    |
| Missing Undergraduate GPA       |           |            |         |             |            |
| Missing TOEFL/IELTS             | -0.2364   | 0.2852     | 0.0796  |             |            |
| Concentration FE                | No        | Yes        | Yes     | Yes         | Yes        |
| Observations                    | 2078      | 2231       | 690     | 737         | 486        |
| Pseudo-R <sup>2</sup>           | 0.2642    | 0.1456     | 0.1589  | 0.1486      | 0.1362     |
| CV Pseudo-R <sup>2</sup>        | 0.2297    | 0.0912     | 0.0592  | 0.0850      | 0.0537     |

- Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

# A4: Simulated Wald Test Results (Same)

Table 19a: Simulated Wald Test Results (Same)

|                                      | Dependent Variable = Accept |                        |                         |                        | Dependent Variable = Success |                        |                         |                        |
|--------------------------------------|-----------------------------|------------------------|-------------------------|------------------------|------------------------------|------------------------|-------------------------|------------------------|
|                                      | Economics                   |                        | Non-Economics           |                        | Economics                    |                        | Non-Economics           |                        |
|                                      | Coef                        | AME                    | Coef                    | AME                    | Coef                         | AME                    | Coef                    | AME                    |
| Intercept                            | -16.3848***<br>(1.1315)     | -1.5822***<br>(0.0956) | -28.5524***<br>(1.4963) | -1.3677***<br>(0.0500) | -10.4586***<br>(0.7236)      | -1.9105***<br>(0.1070) | -10.2610***<br>(0.4815) | -1.8514***<br>(0.0628) |
| GRE Quantitative Percentile          | 0.1509***<br>(0.0131)       | 0.0146***<br>(0.0012)  | 0.3245***<br>(0.0179)   | 0.0155***<br>(0.0007)  | 0.1058***<br>(0.0088)        | 0.0193***<br>(0.0014)  | 0.1154***<br>(0.0070)   | 0.0208***<br>(0.0011)  |
| Gender                               | 0.4714***<br>(0.1475)       | 0.0455***<br>(0.0140)  | 0.7880***<br>(0.1696)   | 0.0377***<br>(0.0081)  | -0.1384<br>(0.1095)          | -0.0253<br>(0.0200)    | -0.0497<br>(0.0895)     | -0.0090<br>(0.0161)    |
| Demographic 1                        | -0.6287***<br>(0.2088)      | -0.0607***<br>(0.0200) | -1.8488***<br>(0.2445)  | -0.0886***<br>(0.0115) | -0.3486**<br>(0.1467)        | -0.0637**<br>(0.0267)  | -0.3079***<br>(0.1123)  | -0.0556***<br>(0.0202) |
| Demographic 2                        | 0.1531<br>(0.2072)          | 0.0148<br>(0.0200)     | 0.3703*<br>(0.1929)     | 0.0177*<br>(0.0091)    | 0.1756<br>(0.1558)           | 0.0321<br>(0.0284)     | 0.0282<br>(0.1069)      | 0.0051<br>(0.0193)     |
| Advanced Math                        | 0.2889*<br>(0.1612)         | 0.0279*<br>(0.0156)    |                         |                        | 0.3267***<br>(0.1112)        | 0.0597***<br>(0.0202)  |                         |                        |
| Committee Score                      | 0.2241***<br>(0.0413)       | 0.0216***<br>(0.0039)  | 0.3994***<br>(0.0506)   | 0.0191***<br>(0.0023)  | 0.1026***<br>(0.0287)        | 0.0187***<br>(0.0052)  | 0.0757***<br>(0.0228)   | 0.0137***<br>(0.0041)  |
| Non-Economics 1                      |                             |                        | 0.8376***<br>(0.2297)   | 0.0401***<br>(0.0107)  |                              |                        | 1.9483***<br>(0.1236)   | 0.3515***<br>(0.0205)  |
| Observations   Pseudo-R <sup>2</sup> | 5000                        | 0.3251                 |                         |                        | 5000                         | 0.1431                 |                         |                        |
| Parameters   LL   AIC                | 14                          | -1130.4                | 2288.8                  |                        | 14                           | -2691.4                | 5410.8                  |                        |
| Wald Statistic   DF   P-Value        | 3.1018                      | 5                      | 0.6843                  |                        | 3.0042                       | 5                      | 0.6993                  |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Wald Test Results (Different)

Table 19b: Simulated Wald Test Results (Different)

|                                      | Dependent Variable = Accept |                        |                        |                        | Dependent Variable = Success |                        |                        |                        |
|--------------------------------------|-----------------------------|------------------------|------------------------|------------------------|------------------------------|------------------------|------------------------|------------------------|
|                                      | Economics                   |                        | Non-Economics          |                        | Economics                    |                        | Non-Economics          |                        |
|                                      | Coef                        | AME                    | Coef                   | AME                    | Coef                         | AME                    | Coef                   | AME                    |
| Intercept                            | -16.3291***<br>(1.1274)     | -1.5784***<br>(0.0955) | -8.4753***<br>(0.8673) | -0.3642***<br>(0.0412) | -10.4581***<br>(0.7237)      | -1.9104***<br>(0.1070) | -3.2236***<br>(0.4018) | -0.5997***<br>(0.0728) |
| GRE Quantitative Percentile          | 0.1502***<br>(0.0131)       | 0.0145***<br>(0.0012)  | 0.0402***<br>(0.0128)  | 0.0017***<br>(0.0006)  | 0.1058***<br>(0.0088)        | 0.0193***<br>(0.0014)  | 0.0034<br>(0.0062)     | 0.0006<br>(0.0012)     |
| Gender                               | 0.4702***<br>(0.1473)       | 0.0454***<br>(0.0140)  | 0.5942***<br>(0.1820)  | 0.0255***<br>(0.0079)  | -0.1372<br>(0.1095)          | -0.0251<br>(0.0200)    | 0.0375<br>(0.0868)     | 0.0070<br>(0.0161)     |
| Demographic 1                        | -0.6246***<br>(0.2085)      | -0.0604***<br>(0.0200) | -0.2541<br>(0.2393)    | -0.0109<br>(0.0103)    | -0.3544**<br>(0.1467)        | -0.0647**<br>(0.0267)  | -0.1152<br>(0.1103)    | -0.0214<br>(0.0205)    |
| Demographic 2                        | 0.1555<br>(0.2070)          | 0.0150<br>(0.0200)     | 0.0081<br>(0.2109)     | 0.0003<br>(0.0091)     | 0.1713<br>(0.1557)           | 0.0313<br>(0.0284)     | -0.0562<br>(0.1030)    | -0.0105<br>(0.0192)    |
| Advanced Math                        | 0.2886*<br>(0.1611)         | 0.0279*<br>(0.0156)    |                        |                        | 0.3284***<br>(0.1112)        | 0.0600***<br>(0.0202)  |                        |                        |
| Committee Score                      | 0.2241***<br>(0.0413)       | 0.0217***<br>(0.0039)  | 0.3817***<br>(0.0528)  | 0.0164***<br>(0.0025)  | 0.1026***<br>(0.0287)        | 0.0187***<br>(0.0052)  | 0.1808***<br>(0.0228)  | 0.0336***<br>(0.0041)  |
| Non-Economics 1                      |                             |                        | -0.0452<br>(0.1995)    | -0.0019<br>(0.0086)    |                              |                        | 1.1393***<br>(0.1219)  | 0.2120***<br>(0.0219)  |
| Observations   Pseudo-R <sup>2</sup> | 5000                        | 0.1488                 |                        |                        | 5000                         | 0.0758                 |                        |                        |
| Parameters   LL   AIC                | 14                          | -1155.7                | 2339.5                 |                        | 14                           | -2747.2                | 5522.4                 |                        |
| Wald Statistic   DF   P-Value        | 112.5897                    | 5                      | 0.0000                 |                        | 124.7379                     | 5                      | 0.0000                 |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Unrestricted Dynamic Estimates (Near-Optimal)

Table 20a: Simulated Unrestricted Dynamic Estimates (Near-Optimal)

|                             | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -1.7577**<br>(0.7805) | -0.4303**<br>(0.1891) | 0.3695<br>(1.4912)    |  | -12.0690***<br>(0.9758) | -2.2483***<br>(0.1337) | -8.3663***<br>(1.7459) | -1.6540***<br>(0.3133) | -7.3172***<br>(1.2885) | -1.0451***<br>(0.1661) |
| GRE Quantitative Percentile | 0.0251***<br>(0.0094) | 0.0061***<br>(0.0023) | 0.0510***<br>(0.0170) |  | 0.1260***<br>(0.0114)   | 0.0235***<br>(0.0017)  | 0.0911***<br>(0.0200)  | 0.0180***<br>(0.0036)  | 0.1028***<br>(0.0142)  | 0.0147***<br>(0.0017)  |
| Gender                      | -0.2435*<br>(0.1312)  | -0.0596*<br>(0.0319)  | 0.1807<br>(0.2034)    |  | 0.0400<br>(0.1525)      | 0.0074<br>(0.0284)     | 0.0150<br>(0.2415)     | 0.0030<br>(0.0477)     | 0.2259<br>(0.1762)     | 0.0323<br>(0.0251)     |
| Demographic 1               | -0.0504<br>(0.1730)   | -0.0123<br>(0.0424)   | -0.6989**<br>(0.2889) |  | 0.0163<br>(0.2031)      | 0.0030<br>(0.0378)     | -0.0503<br>(0.3722)    | -0.0100<br>(0.0736)    | -0.4384*<br>(0.2358)   | -0.0626*<br>(0.0334)   |
| Demographic 2               | -0.3121<br>(0.1930)   | -0.0764<br>(0.0470)   | -0.4226<br>(0.3254)   |  | 0.1678<br>(0.2315)      | 0.0313<br>(0.0431)     | -0.2878<br>(0.4145)    | -0.0569<br>(0.0818)    | -0.4682*<br>(0.2734)   | -0.0669*<br>(0.0387)   |
| Advanced Math               |                       |                       | 1.1347***<br>(0.1992) |  |                         |                        | 0.2066<br>(0.2496)     | 0.0408<br>(0.0491)     | 0.7127***<br>(0.1791)  | 0.1018***<br>(0.0247)  |
| Committee Score             |                       |                       |                       |  |                         |                        | 0.0600<br>(0.0639)     | 0.0119<br>(0.0126)     | -0.0790<br>(0.0688)    | -0.0113<br>(0.0098)    |
| Missing Committee Score     |                       |                       |                       |  |                         |                        |                        |                        | 0.0486<br>(0.1969)     | 0.0069<br>(0.0281)     |
| Year Transition More        | -0.2523**<br>(0.1282) | -0.0618**<br>(0.0311) | 0.1483<br>(0.2025)    |  | 1.1441***<br>(0.1501)   | 0.2131***<br>(0.0250)  | -1.1114***<br>(0.2446) | -0.2197***<br>(0.0435) | -0.3543**<br>(0.1794)  | -0.0506**<br>(0.0254)  |
| Program Rank                |                       |                       |                       |  |                         |                        |                        |                        | -0.0302***<br>(0.0018) | -0.0043***<br>(0.0002) |
| $\sqrt{\text{MSE}}$         |                       |                       | 1.8393***<br>(0.0614) |  |                         |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC      | 1000                  | 38                    | -2602.7               |  | 5281.5                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

|                             | Advanced Math         |                       | Committee Score        |  | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|------------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   | Coefficient            |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -1.7577**<br>(0.7805) | -0.4303**<br>(0.1891) | 1.2187<br>(1.4016)     |  | -10.3454***<br>(0.9474) | -2.0151***<br>(0.1440) | -4.8695***<br>(1.5490) | -1.0694***<br>(0.3252) | -7.3017***<br>(1.2721) | -1.0448***<br>(0.1635) |
| GRE Quantitative Percentile | 0.0251***<br>(0.0094) | 0.0061***<br>(0.0023) | 0.0431***<br>(0.0164)  |  | 0.1103***<br>(0.0111)   | 0.0215***<br>(0.0018)  | 0.0488***<br>(0.0180)  | 0.0107***<br>(0.0038)  | 0.1021***<br>(0.0140)  | 0.0146***<br>(0.0017)  |
| Gender                      | -0.2435*<br>(0.1312)  | -0.0596*<br>(0.0319)  | 0.1707<br>(0.2037)     |  | -0.0472<br>(0.1485)     | -0.0092<br>(0.0289)    | 0.1746<br>(0.2312)     | 0.0383<br>(0.0506)     | 0.2087<br>(0.1755)     | 0.0299<br>(0.0251)     |
| Demographic 1               | -0.0504<br>(0.1730)   | -0.0123<br>(0.0424)   | -0.7422***<br>(0.2732) |  | -0.4655**<br>(0.1978)   | -0.0907**<br>(0.0382)  | -0.5625*<br>(0.3192)   | -0.1235*<br>(0.0689)   | -0.4120*<br>(0.2351)   | -0.0590*<br>(0.0334)   |
| Demographic 2               | -0.3121<br>(0.1930)   | -0.0764<br>(0.0470)   | -0.4868*<br>(0.2925)   |  | 0.1277<br>(0.2186)      | 0.0249<br>(0.0426)     | -0.1757<br>(0.3381)    | -0.0386<br>(0.0742)    | -0.3798<br>(0.2696)    | -0.0543<br>(0.0384)    |
| Advanced Math               |                       |                       | 1.1482***<br>(0.1999)  |  |                         |                        | -0.0098<br>(0.2366)    | -0.0021<br>(0.0520)    | 0.7178***<br>(0.1795)  | 0.1027***<br>(0.0248)  |
| Committee Score             |                       |                       |                        |  |                         |                        | 0.1714***<br>(0.0604)  | 0.0376***<br>(0.0128)  | -0.0769<br>(0.0684)    | -0.0110<br>(0.0098)    |
| Missing Committee Score     |                       |                       |                        |  |                         |                        |                        |                        | 0.1129<br>(0.1909)     | 0.0162<br>(0.0273)     |
| Year Transition More        | -0.2523**<br>(0.1282) | -0.0618**<br>(0.0311) | 0.0871<br>(0.2014)     |  | 1.0879***<br>(0.1466)   | 0.2119***<br>(0.0255)  | -0.7286***<br>(0.2324) | -0.1600***<br>(0.0485) | -0.3741**<br>(0.1781)  | -0.0535**<br>(0.0253)  |
| Program Rank                |                       |                       |                        |  |                         |                        |                        |                        | -0.0309***<br>(0.0019) | -0.0044***<br>(0.0002) |
| $\sqrt{\text{MSE}}$         |                       |                       | 1.8362***<br>(0.0603)  |  |                         |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC      | 1000                  | 38                    | -2648.4                |  | 5372.9                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Restricted Dynamic Estimates (Near-Optimal)

Table 21a: Simulated Restricted Dynamic Estimates (Near-Optimal)

|                             | Advanced Math         | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|-----------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -1.9394**<br>(0.7791) | 0.3771<br>(1.4942)    | -0.1457**<br>(0.0663)  | 0.8644<br>(0.7591, 0.9843) | 0.2458<br>(0.2300) | 56.11%<br>(44.89%, 66.74%) | -7.8112***<br>(1.1221) |                       |
| GRE Quantitative Percentile | 0.0273***<br>(0.0094) | 0.0509***<br>(0.0171) |                        |                            |                    |                            | 0.1044***<br>(0.0128)  |                       |
| Gender                      | -0.2478*<br>(0.1310)  | 0.1808<br>(0.2034)    |                        |                            |                    |                            | 0.0937<br>(0.1163)     |                       |
| Demographic 1               | -0.0432<br>(0.1727)   | -0.6993**<br>(0.2904) |                        |                            |                    |                            | -0.2023<br>(0.1559)    |                       |
| Demographic 2               | -0.3026<br>(0.1930)   | -0.4231<br>(0.3260)   |                        |                            |                    |                            | -0.2735<br>(0.1817)    |                       |
| Advanced Math               |                       | 1.1347***<br>(0.1992) |                        |                            |                    |                            | 0.4597***<br>(0.1452)  |                       |
| Committee Score             |                       |                       |                        |                            |                    |                            | -0.0021<br>(0.0436)    |                       |
| Missing Committee Score     |                       |                       |                        |                            |                    |                            | 0.0240<br>(0.1891)     |                       |
| Year Transition More        | -0.2448*<br>(0.1281)  | 0.1481<br>(0.2023)    | -0.1684***<br>(0.0637) | 0.7304<br>(0.6462, 0.8256) | 0.4096<br>(0.2744) | 65.82%<br>(57.07%, 73.62%) | -0.3690**<br>(0.1763)  |                       |
| Program Rank                |                       |                       |                        |                            |                    |                            | -0.0300***<br>(0.0018) |                       |
| $\sqrt{\text{MSE}}$         |                       | 1.8393***<br>(0.0614) |                        |                            |                    |                            |                        |                       |
| Sigma                       |                       |                       |                        |                            |                    |                            |                        | 0.1647***<br>(0.0242) |
| Obs   Param   LL   AIC      | 1000                  | 29                    | -2642.8                | 5343.7                     |                    |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 21b: Simulated Restricted Dynamic Estimates (Suboptimal)

|                             | Advanced Math         | Committee Score        | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|-----------------------|------------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | -1.9242**<br>(0.7773) | 1.0267<br>(1.4057)     | -0.0633<br>(0.0892)    | 0.9387<br>(0.7882, 1.1179) | 0.1263<br>(0.2508) | 53.15%<br>(40.97%, 64.97%) | -7.7219***<br>(1.1783) |                       |
| GRE Quantitative Percentile | 0.0271***<br>(0.0093) | 0.0453***<br>(0.0165)  |                        |                            |                    |                            | 0.1006***<br>(0.0134)  |                       |
| Gender                      | -0.2513*<br>(0.1310)  | 0.1636<br>(0.2039)     |                        |                            |                    |                            | 0.1366<br>(0.1297)     |                       |
| Demographic 1               | -0.0530<br>(0.1728)   | -0.7462***<br>(0.2730) |                        |                            |                    |                            | -0.5186***<br>(0.1682) |                       |
| Demographic 2               | -0.3004<br>(0.1928)   | -0.4758<br>(0.2928)    |                        |                            |                    |                            | -0.2123<br>(0.1992)    |                       |
| Advanced Math               |                       | 1.1476***<br>(0.1999)  |                        |                            |                    |                            | 0.4443***<br>(0.1595)  |                       |
| Committee Score             |                       |                        |                        |                            |                    |                            | 0.0635<br>(0.0480)     |                       |
| Missing Committee Score     |                       |                        |                        |                            |                    |                            | 0.1011<br>(0.1862)     |                       |
| Year Transition More        | -0.2466*<br>(0.1281)  | 0.0910<br>(0.2010)     | -0.2559***<br>(0.0710) | 0.7268<br>(0.6177, 0.8551) | 0.3083<br>(0.3058) | 60.70%<br>(50.39%, 70.13%) | -0.3596**<br>(0.1749)  |                       |
| Program Rank                |                       |                        |                        |                            |                    |                            | -0.0303***<br>(0.0018) |                       |
| $\sqrt{\text{MSE}}$         |                       | 1.8362***<br>(0.0603)  |                        |                            |                    |                            |                        |                       |
| Sigma                       |                       |                        |                        |                            |                    |                            |                        | 0.2258***<br>(0.0411) |
| Obs   Param   LL   AIC      | 1000                  | 29                     | -2686.1                | 5430.2                     |                    |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Unrestricted Static Estimates (Near-Optimal)

Table 22a: Simulated Unrestricted Static Estimates (Near-Optimal)

|                             | Transition              |                        | Accept                 |                        | Success 1              |                        | Success 2              |                        |
|-----------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -12.0690***<br>(0.9758) | -2.2483***<br>(0.1337) | -8.3663***<br>(1.7459) | -1.6540***<br>(0.3133) | -7.3875***<br>(1.1197) | -1.0821***<br>(0.1406) | -7.3172***<br>(1.2885) | -1.0451***<br>(0.1661) |
| GRE Quantitative Percentile | 0.1260***<br>(0.0114)   | 0.0235***<br>(0.0017)  | 0.0911***<br>(0.0200)  | 0.0180***<br>(0.0036)  | 0.1043***<br>(0.0134)  | 0.0153***<br>(0.0016)  | 0.1028***<br>(0.0142)  | 0.0147***<br>(0.0017)  |
| Gender                      | 0.0400<br>(0.1525)      | 0.0074<br>(0.0284)     | 0.0150<br>(0.2415)     | 0.0030<br>(0.0477)     | 0.1580<br>(0.1732)     | 0.0231<br>(0.0254)     | 0.2259<br>(0.1762)     | 0.0323<br>(0.0251)     |
| Demographic 1               | 0.0163<br>(0.2031)      | 0.0030<br>(0.0378)     | -0.0503<br>(0.3722)    | -0.0100<br>(0.0736)    | -0.4556**<br>(0.2269)  | -0.0667**<br>(0.0328)  | -0.4384*<br>(0.2358)   | -0.0626*<br>(0.0334)   |
| Demographic 2               | 0.1678<br>(0.2315)      | 0.0313<br>(0.0431)     | -0.2878<br>(0.4145)    | -0.0569<br>(0.0818)    | -0.5153*<br>(0.2641)   | -0.0755**<br>(0.0383)  | -0.4682*<br>(0.2734)   | -0.0669*<br>(0.0387)   |
| Advanced Math               |                         |                        | 0.2066<br>(0.2496)     | 0.0408<br>(0.0491)     |                        |                        | 0.7127***<br>(0.1791)  | 0.1018***<br>(0.0247)  |
| Committee Score             |                         |                        | 0.0600<br>(0.0639)     | 0.0119<br>(0.0126)     |                        |                        | -0.0790<br>(0.0688)    | -0.0113<br>(0.0098)    |
| Missing Committee Score     |                         |                        |                        |                        |                        |                        | 0.0486<br>(0.1969)     | 0.0069<br>(0.0281)     |
| Year Transition More        | 1.1441***<br>(0.1501)   | 0.2131***<br>(0.0250)  | -1.1114***<br>(0.2446) | -0.2197***<br>(0.0435) | -0.3942**<br>(0.1689)  | -0.0577**<br>(0.0244)  | -0.3543**<br>(0.1794)  | -0.0506**<br>(0.0254)  |
| Program Rank                |                         |                        |                        |                        | -0.0299***<br>(0.0018) | -0.0044***<br>(0.0002) | -0.0302***<br>(0.0018) | -0.0043***<br>(0.0002) |
| Obs   Param   LL   AIC      | 1000                    | 31                     | -1632.5                | 3327.1                 |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Unrestricted Static Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Static Estimates (Suboptimal)

|                             | Transition              |                        | Accept                 |                        | Success 1              |                        | Success 2              |                        |
|-----------------------------|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | -10.3454***<br>(0.9474) | -2.0151***<br>(0.1440) | -4.8695***<br>(1.5490) | -1.0694***<br>(0.3252) | -7.2362***<br>(1.1209) | -1.0632***<br>(0.1418) | -7.3017***<br>(1.2721) | -1.0448***<br>(0.1635) |
| GRE Quantitative Percentile | 0.1103***<br>(0.0111)   | 0.0215***<br>(0.0018)  | 0.0488***<br>(0.0180)  | 0.0107***<br>(0.0038)  | 0.1026***<br>(0.0134)  | 0.0151***<br>(0.0016)  | 0.1021***<br>(0.0140)  | 0.0146***<br>(0.0017)  |
| Gender                      | -0.0472<br>(0.1485)     | -0.0092<br>(0.0289)    | 0.1746<br>(0.2312)     | 0.0383<br>(0.0506)     | 0.1408<br>(0.1727)     | 0.0207<br>(0.0254)     | 0.2087<br>(0.1755)     | 0.0299<br>(0.0251)     |
| Demographic 1               | -0.4655**<br>(0.1978)   | -0.0907**<br>(0.0382)  | -0.5625*<br>(0.3192)   | -0.1235*<br>(0.0689)   | -0.4199*<br>(0.2285)   | -0.0617*<br>(0.0332)   | -0.4120*<br>(0.2351)   | -0.0590*<br>(0.0334)   |
| Demographic 2               | 0.1277<br>(0.2186)      | 0.0249<br>(0.0426)     | -0.1757<br>(0.3381)    | -0.0386<br>(0.0742)    | -0.4343*<br>(0.2605)   | -0.0638*<br>(0.0380)   | -0.3798<br>(0.2696)    | -0.0543<br>(0.0384)    |
| Advanced Math               |                         |                        | -0.0098<br>(0.2366)    | -0.0021<br>(0.0520)    |                        |                        | 0.7178***<br>(0.1795)  | 0.1027***<br>(0.0248)  |
| Committee Score             |                         |                        | 0.1714***<br>(0.0604)  | 0.0376***<br>(0.0128)  |                        |                        | -0.0769<br>(0.0684)    | -0.0110<br>(0.0098)    |
| Missing Committee Score     |                         |                        |                        |                        |                        |                        | 0.1129<br>(0.1909)     | 0.0162<br>(0.0273)     |
| Year Transition More        | 1.0879***<br>(0.1466)   | 0.2119***<br>(0.0255)  | -0.7286***<br>(0.2324) | -0.1600***<br>(0.0485) | -0.4350**<br>(0.1689)  | -0.0639***<br>(0.0244) | -0.3741**<br>(0.1781)  | -0.0535**<br>(0.0253)  |
| Program Rank                |                         |                        |                        |                        | -0.0305***<br>(0.0019) | -0.0045***<br>(0.0002) | -0.0309***<br>(0.0019) | -0.0044***<br>(0.0002) |
| Obs   Param   LL   AIC      | 1000                    | 31                     | -1673.0                | 3408.1                 |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Restricted Static Estimates (Near-Optimal)

Table 23a: Simulated Restricted Static Estimates (Near-Optimal)

|                             | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2            | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Sigma                 |
|-----------------------------|------------------------|----------------------------|---------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                   | 1.2114***<br>(0.2502)  | 77.05%<br>(67.28%, 84.58%) | 0.3765<br>(0.2666)  | 59.30%<br>(46.35%, 71.08%) | -7.7989***<br>(1.0685) | -6.9427***<br>(1.2241) |                       |
| GRE Quantitative Percentile |                        |                            |                     |                            | 0.1065***<br>(0.0128)  | 0.0957***<br>(0.0139)  |                       |
| Gender                      |                        |                            |                     |                            | 0.0694<br>(0.1077)     | 0.1374<br>(0.1373)     |                       |
| Demographic 1               |                        |                            |                     |                            | -0.1638<br>(0.1410)    | -0.3053<br>(0.1944)    |                       |
| Demographic 2               |                        |                            |                     |                            | -0.0994<br>(0.1623)    | -0.3969*<br>(0.2189)   |                       |
| Advanced Math               |                        |                            |                     |                            |                        | 0.5127***<br>(0.1456)  |                       |
| Committee Score             |                        |                            |                     |                            |                        | -0.0104<br>(0.0458)    |                       |
| Missing Committee Score     |                        |                            |                     |                            |                        | -0.0145<br>(0.1924)    |                       |
| Year Transition More        | -1.3691***<br>(0.2738) | 46.07%<br>(40.01%, 52.24%) | 0.6127*<br>(0.3291) | 72.89%<br>(61.87%, 81.67%) | -0.3712**<br>(0.1649)  | -0.3819**<br>(0.1814)  |                       |
| Program Rank                |                        |                            |                     |                            | -0.0298***<br>(0.0018) | -0.0300***<br>(0.0018) |                       |
| Sigma                       |                        |                            |                     |                            |                        |                        | 0.2021***<br>(0.0252) |
| Obs   Param   LL   AIC      | 1000                   | 22                         | -1639.7             | 3323.5                     |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Restricted Static Estimates (Suboptimal)

**Table 23b: Simulated Restricted Static Estimates (Suboptimal)**

|                             | Cutoff 1               | $P(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success 1              | Success 2              | Sigma                 |
|-----------------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                   | 1.5071***<br>(0.3814)  | 81.86%<br>(68.13%, 90.50%) | 0.2881<br>(0.2858) | 57.15%<br>(43.24%, 70.02%) | -7.9222***<br>(1.1110) | -6.4145***<br>(1.2821) |                       |
| GRE Quantitative Percentile |                        |                            |                    |                            | 0.1102***<br>(0.0132)  | 0.0864***<br>(0.0144)  |                       |
| Gender                      |                        |                            |                    |                            | 0.0340<br>(0.1208)     | 0.2000<br>(0.1433)     |                       |
| Demographic 1               |                        |                            |                    |                            | -0.4496***<br>(0.1586) | -0.5033***<br>(0.1917) |                       |
| Demographic 2               |                        |                            |                    |                            | -0.1170<br>(0.1795)    | -0.3225<br>(0.2194)    |                       |
| Advanced Math               |                        |                            |                    |                            |                        | 0.4844***<br>(0.1548)  |                       |
| Committee Score             |                        |                            |                    |                            |                        | 0.0524<br>(0.0491)     |                       |
| Missing Committee Score     |                        |                            |                    |                            |                        | 0.0419<br>(0.1918)     |                       |
| Year Transition More        | -1.6799***<br>(0.3946) | 45.69%<br>(39.03%, 52.51%) | 0.3660<br>(0.3462) | 65.79%<br>(53.86%, 76.01%) | -0.4035**<br>(0.1655)  | -0.3931**<br>(0.1786)  |                       |
| Program Rank                |                        |                            |                    |                            | -0.0303***<br>(0.0019) | -0.0304***<br>(0.0018) |                       |
| Sigma                       |                        |                            |                    |                            |                        |                        | 0.2573***<br>(0.0392) |
| Obs   Param   LL   AIC      | 1000                   | 22                         | -1687.8            | 3419.7                     |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Unrestricted Myopic Estimates (Near-Optimal)

Table 24a: Simulated Unrestricted Myopic Estimates (Near-Optimal)

|                             | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | 0.1765**<br>(0.0898)  | 0.0439**<br>(0.0222)  | 4.4154***<br>(0.1909) |  | -12.0690***<br>(0.9758) | -2.2483***<br>(0.1337) | -8.3663***<br>(1.7459) | -1.6540***<br>(0.3133) | -7.3172***<br>(1.2885) | -1.0451***<br>(0.1661) |
| GRE Quantitative Percentile |                       |                       |                       |  | 0.1260***<br>(0.0114)   | 0.0235***<br>(0.0017)  | 0.0911***<br>(0.0200)  | 0.0180***<br>(0.0036)  | 0.1028***<br>(0.0142)  | 0.0147***<br>(0.0017)  |
| Gender                      |                       |                       |                       |  | 0.0400<br>(0.1525)      | 0.0074<br>(0.0284)     | 0.0150<br>(0.2415)     | 0.0030<br>(0.0477)     | 0.2259<br>(0.1762)     | 0.0323<br>(0.0251)     |
| Demographic 1               |                       |                       |                       |  | 0.0163<br>(0.2031)      | 0.0030<br>(0.0378)     | -0.0503<br>(0.3722)    | -0.0100<br>(0.0736)    | -0.4384*<br>(0.2358)   | -0.0626*<br>(0.0334)   |
| Demographic 2               |                       |                       |                       |  | 0.1678<br>(0.2315)      | 0.0313<br>(0.0431)     | -0.2878<br>(0.4145)    | -0.0569<br>(0.0818)    | -0.4682*<br>(0.2734)   | -0.0669*<br>(0.0387)   |
| Advanced Math               |                       |                       | 1.1642***<br>(0.1981) |  |                         |                        | 0.2066<br>(0.2496)     | 0.0408<br>(0.0491)     | 0.7127***<br>(0.1791)  | 0.1018***<br>(0.0247)  |
| Committee Score             |                       |                       |                       |  |                         |                        | 0.0600<br>(0.0639)     | 0.0119<br>(0.0126)     | -0.0790<br>(0.0688)    | -0.0113<br>(0.0098)    |
| Missing Committee Score     |                       |                       |                       |  |                         |                        |                        |                        | 0.0486<br>(0.1969)     | 0.0069<br>(0.0281)     |
| Year Transition More        | -0.2485**<br>(0.1268) | -0.0618**<br>(0.0313) | 0.1348<br>(0.2040)    |  | 1.1441***<br>(0.1501)   | 0.2131***<br>(0.0250)  | -1.1114***<br>(0.2446) | -0.2197***<br>(0.0435) | -0.3543**<br>(0.1794)  | -0.0506**<br>(0.0254)  |
| Program Rank                |                       |                       |                       |  |                         |                        |                        |                        | -0.0302***<br>(0.0018) | -0.0043***<br>(0.0002) |
| $\sqrt{MSE}$                |                       |                       | 1.8730***<br>(0.0618) |  |                         |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC      | 1000                  | 30                    | -2617.5               |  | 5295.0                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Myopic Estimates (Suboptimal)

|                             | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                 |                        | Success                |                        |
|-----------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                             | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                   | 0.1765**<br>(0.0898)  | 0.0439**<br>(0.0222)  | 4.5276***<br>(0.1915) |  | -10.3454***<br>(0.9474) | -2.0151***<br>(0.1440) | -4.8695***<br>(1.5490) | -1.0694***<br>(0.3252) | -7.3017***<br>(1.2721) | -1.0448***<br>(0.1635) |
| GRE Quantitative Percentile |                       |                       |                       |  | 0.1103***<br>(0.0111)   | 0.0215***<br>(0.0018)  | 0.0488***<br>(0.0180)  | 0.0107***<br>(0.0038)  | 0.1021***<br>(0.0140)  | 0.0146***<br>(0.0017)  |
| Gender                      |                       |                       |                       |  | -0.0472<br>(0.1485)     | -0.0092<br>(0.0289)    | 0.1746<br>(0.2312)     | 0.0383<br>(0.0506)     | 0.2087<br>(0.1755)     | 0.0299<br>(0.0251)     |
| Demographic 1               |                       |                       |                       |  | -0.4655**<br>(0.1978)   | -0.0907**<br>(0.0382)  | -0.5625*<br>(0.3192)   | -0.1235*<br>(0.0689)   | -0.4120*<br>(0.2351)   | -0.0590*<br>(0.0334)   |
| Demographic 2               |                       |                       |                       |  | 0.1277<br>(0.2186)      | 0.0249<br>(0.0426)     | -0.1757<br>(0.3381)    | -0.0386<br>(0.0742)    | -0.3798<br>(0.2696)    | -0.0543<br>(0.0384)    |
| Advanced Math               |                       |                       | 1.1789***<br>(0.1973) |  |                         |                        | -0.0098<br>(0.2366)    | -0.0021<br>(0.0520)    | 0.7178***<br>(0.1795)  | 0.1027***<br>(0.0248)  |
| Committee Score             |                       |                       |                       |  |                         |                        | 0.1714***<br>(0.0604)  | 0.0376***<br>(0.0128)  | -0.0769<br>(0.0684)    | -0.0110<br>(0.0098)    |
| Missing Committee Score     |                       |                       |                       |  |                         |                        |                        |                        | 0.1129<br>(0.1909)     | 0.0162<br>(0.0273)     |
| Year Transition More        | -0.2485**<br>(0.1268) | -0.0618**<br>(0.0313) | 0.0712<br>(0.2044)    |  | 1.0879***<br>(0.1466)   | 0.2119***<br>(0.0255)  | -0.7286***<br>(0.2324) | -0.1600***<br>(0.0485) | -0.3741**<br>(0.1781)  | -0.0535**<br>(0.0253)  |
| Program Rank                |                       |                       |                       |  |                         |                        |                        |                        | -0.0309***<br>(0.0019) | -0.0044***<br>(0.0002) |
| $\sqrt{MSE}$                |                       |                       | 1.8655***<br>(0.0600) |  |                         |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC      | 1000                  | 30                    | -2662.4               |  | 5384.8                  |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \*p<0.10, \*\*p<0.05, \*\*\*p<0.01.

# A4: Simulated Restricted Myopic Estimates (Near-Optimal)

Table 25a: Simulated Restricted Myopic Estimates (Near-Optimal)

|                             | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|----------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | 0.1822**<br>(0.0898) | 4.4131***<br>(0.1915) | -0.1459**<br>(0.0669)  | 0.8642<br>(0.7581, 0.9852) | 0.2450<br>(0.2303) | 56.09%<br>(44.86%, 66.74%) | -7.8232***<br>(1.1263) |                       |
| GRE Quantitative Percentile |                      |                       |                        |                            |                    |                            | 0.1055***<br>(0.0128)  |                       |
| Gender                      |                      |                       |                        |                            |                    |                            | 0.0872<br>(0.1160)     |                       |
| Demographic 1               |                      |                       |                        |                            |                    |                            | -0.2089<br>(0.1545)    |                       |
| Demographic 2               |                      |                       |                        |                            |                    |                            | -0.2850<br>(0.1804)    |                       |
| Advanced Math               |                      | 1.1642***<br>(0.1982) |                        |                            |                    |                            | 0.4385***<br>(0.1452)  |                       |
| Committee Score             |                      |                       |                        |                            |                    |                            | -0.0125<br>(0.0436)    |                       |
| Missing Committee Score     |                      |                       |                        |                            |                    |                            | 0.0243<br>(0.1889)     |                       |
| Year Transition More        | -0.2418*<br>(0.1267) | 0.1344<br>(0.2039)    | -0.1671***<br>(0.0640) | 0.7312<br>(0.6463, 0.8273) | 0.4173<br>(0.2757) | 65.98%<br>(57.16%, 73.81%) | -0.3689**<br>(0.1764)  |                       |
| Program Rank                |                      |                       |                        |                            |                    |                            | -0.0300***<br>(0.0018) |                       |
| $\sqrt{\text{MSE}}$         |                      | 1.8730***<br>(0.0618) |                        |                            |                    |                            |                        |                       |
| Sigma                       |                      |                       |                        |                            |                    |                            |                        | 0.1652***<br>(0.0244) |
| Obs   Param   LL   AIC      | 1000                 | 21                    | -2658.4                | 5358.7                     |                    |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Restricted Myopic Estimates (Suboptimal)

Table 25b: Simulated Restricted Myopic Estimates (Suboptimal)

|                             | Advanced Math        | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Sigma                 |
|-----------------------------|----------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|-----------------------|
| Intercept                   | 0.1801**<br>(0.0898) | 4.5328***<br>(0.1919) | -0.0597<br>(0.0912)    | 0.9420<br>(0.7878, 1.1264) | 0.1255<br>(0.2525) | 53.13%<br>(40.87%, 65.03%) | -7.7935***<br>(1.1728) |                       |
| GRE Quantitative Percentile |                      |                       |                        |                            |                    |                            | 0.1025***<br>(0.0132)  |                       |
| Gender                      |                      |                       |                        |                            |                    |                            | 0.1327<br>(0.1299)     |                       |
| Demographic 1               |                      |                       |                        |                            |                    |                            | -0.5335***<br>(0.1674) |                       |
| Demographic 2               |                      |                       |                        |                            |                    |                            | -0.2291<br>(0.1982)    |                       |
| Advanced Math               |                      | 1.1784***<br>(0.1973) |                        |                            |                    |                            | 0.4297***<br>(0.1610)  |                       |
| Committee Score             |                      |                       |                        |                            |                    |                            | 0.0501<br>(0.0488)     |                       |
| Missing Committee Score     |                      |                       |                        |                            |                    |                            | 0.1041<br>(0.1858)     |                       |
| Year Transition More        | -0.2436*<br>(0.1267) | 0.0726<br>(0.2043)    | -0.2565***<br>(0.0714) | 0.7289<br>(0.6175, 0.8605) | 0.3183<br>(0.3098) | 60.92%<br>(50.44%, 70.47%) | -0.3616**<br>(0.1752)  |                       |
| Program Rank                |                      |                       |                        |                            |                    |                            | -0.0303***<br>(0.0018) |                       |
| $\sqrt{\text{MSE}}$         |                      | 1.8654***<br>(0.0599) |                        |                            |                    |                            |                        |                       |
| Sigma                       |                      |                       |                        |                            |                    |                            |                        | 0.2283***<br>(0.0424) |
| Obs   Param   LL   AIC      | 1000                 | 21                    | -2701.0                | 5444.0                     |                    |                            |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

# A4: Simulated Unrestricted Matriculation-Dynamic Estimates (Near-Optimal)

**Table 26a: Simulated Unrestricted Matriculation-Dynamic Estimates (Near-Optimal)**

|                                  | Advanced Math         |                       | Committee Score       |  | Transition              |                        | Accept                 |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|-----------------------|-----------------------|-----------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient           | AME                   | Coefficient           |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -1.7577**<br>(0.7805) | -0.4303**<br>(0.1891) | 0.3695<br>(1.4912)    |  | -12.0690***<br>(0.9758) | -2.2483***<br>(0.1337) | -8.3663***<br>(1.7459) | -1.6540***<br>(0.3133) | -7.3172***<br>(1.2885) | -1.0451***<br>(0.1661) | 10.8487*<br>(5.7339)   | 1.1240**<br>(0.5653)   |
| GRE Quantitative Percentile      | 0.0251***<br>(0.0094) | 0.0061***<br>(0.0023) | 0.0510***<br>(0.0170) |  | 0.1260***<br>(0.0114)   | 0.0235***<br>(0.0017)  | 0.0911***<br>(0.0200)  | 0.0180***<br>(0.0036)  | 0.1028***<br>(0.0142)  | 0.0147***<br>(0.0017)  | -0.1049*<br>(0.0630)   | -0.0109*<br>(0.0063)   |
| Gender                           | -0.2435*<br>(0.1312)  | -0.0596*<br>(0.0319)  | 0.1807<br>(0.2034)    |  | 0.0400<br>(0.1525)      | 0.0074<br>(0.0284)     | 0.0150<br>(0.2415)     | 0.0030<br>(0.0477)     | 0.2259<br>(0.1762)     | 0.0323<br>(0.0251)     | -0.9225<br>(0.5834)    | -0.0956*<br>(0.0575)   |
| Demographic 1                    | -0.0504<br>(0.1730)   | -0.0123<br>(0.0424)   | -0.6989**<br>(0.2889) |  | 0.0163<br>(0.2031)      | 0.0030<br>(0.0378)     | -0.0503<br>(0.3722)    | -0.0100<br>(0.0736)    | -0.4384*<br>(0.2358)   | -0.0626*<br>(0.0334)   | 1.2693<br>(0.8109)     | 0.1315<br>(0.0828)     |
| Demographic 2                    | -0.3121<br>(0.1930)   | -0.0764<br>(0.0470)   | -0.4226<br>(0.3254)   |  | 0.1678<br>(0.2315)      | 0.0313<br>(0.0431)     | -0.2878<br>(0.4145)    | -0.0569<br>(0.0818)    | -0.4682*<br>(0.2734)   | -0.0669*<br>(0.0387)   | 0.2958<br>(0.9734)     | 0.0306<br>(0.1014)     |
| Advanced Math                    |                       |                       | 1.1347***<br>(0.1992) |  |                         |                        | 0.2066<br>(0.2496)     | 0.0408<br>(0.0491)     | 0.7127***<br>(0.1791)  | 0.1018***<br>(0.0247)  | 0.3819<br>(0.5414)     | 0.0396<br>(0.0557)     |
| Committee Score                  |                       |                       |                       |  |                         |                        | 0.0600<br>(0.0639)     | 0.0119<br>(0.0126)     | -0.0790<br>(0.0688)    | -0.0113<br>(0.0098)    | -1.0758***<br>(0.2376) | -0.1115***<br>(0.0153) |
| Missing Committee Score          |                       |                       |                       |  |                         |                        |                        |                        | 0.0486<br>(0.1969)     | 0.0069<br>(0.0281)     |                        |                        |
| Year Transition/Matriculate More | -0.2523**<br>(0.1282) | -0.0618**<br>(0.0311) | 0.1483<br>(0.2025)    |  | 1.1441***<br>(0.1501)   | 0.2131***<br>(0.0250)  | -1.1114***<br>(0.2446) | -0.2197***<br>(0.0435) | -0.3543**<br>(0.1794)  | -0.0506**<br>(0.0254)  | 5.2621***<br>(0.8695)  | 0.5452***<br>(0.0260)  |
| Program Rank                     |                       |                       |                       |  |                         |                        |                        |                        | -0.0302***<br>(0.0018) | -0.0043***<br>(0.0002) |                        |                        |
| $\sqrt{\text{MSE}}$              |                       |                       | 1.8393***<br>(0.0614) |  |                         |                        |                        |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC           | 1000                  | 46                    | -2644.2               |  | 5380.5                  |                        |                        |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

# A4: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

**Table 26b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)**

|                                  | Advanced Math         |                       | Committee Score        |  | Transition              |                        | Accept                 |                        | Success                |                        | Matriculate            |                        |
|----------------------------------|-----------------------|-----------------------|------------------------|--|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                                  | Coefficient           | AME                   | Coefficient            |  | Coefficient             | AME                    | Coefficient            | AME                    | Coefficient            | AME                    | Coefficient            | AME                    |
| Intercept                        | -1.7577**<br>(0.7805) | -0.4303**<br>(0.1891) | 1.2187<br>(1.4016)     |  | -10.3454***<br>(0.9474) | -2.0151***<br>(0.1440) | -4.8695***<br>(1.5490) | -1.0694***<br>(0.3252) | -7.3017***<br>(1.2721) | -1.0448***<br>(0.1635) | 5.7452<br>(4.2604)     | 0.5714<br>(0.4164)     |
| GRE Quantitative Percentile      | 0.0251***<br>(0.0094) | 0.0061***<br>(0.0023) | 0.0431***<br>(0.0164)  |  | 0.1103***<br>(0.0111)   | 0.0215***<br>(0.0018)  | 0.0488***<br>(0.0180)  | 0.0107***<br>(0.0038)  | 0.1021***<br>(0.0140)  | 0.0146***<br>(0.0017)  | -0.0443<br>(0.0468)    | -0.0044<br>(0.0046)    |
| Gender                           | -0.2435*<br>(0.1312)  | -0.0596*<br>(0.0319)  | 0.1707<br>(0.2037)     |  | -0.0472<br>(0.1485)     | -0.0092<br>(0.0289)    | 0.1746<br>(0.2312)     | 0.0383<br>(0.0506)     | 0.2087<br>(0.1755)     | 0.0299<br>(0.0251)     | -1.0062*<br>(0.5693)   | -0.1001*<br>(0.0535)   |
| Demographic 1                    | -0.0504<br>(0.1730)   | -0.0123<br>(0.0424)   | -0.7422***<br>(0.2732) |  | -0.4655**<br>(0.1978)   | -0.0907**<br>(0.0382)  | -0.5625*<br>(0.3192)   | -0.1235*<br>(0.0689)   | -0.4120*<br>(0.2351)   | -0.0590*<br>(0.0334)   | 0.8111<br>(0.7062)     | 0.0807<br>(0.0712)     |
| Demographic 2                    | -0.3121<br>(0.1930)   | -0.0764<br>(0.0470)   | -0.4868*<br>(0.2925)   |  | 0.1277<br>(0.2186)      | 0.0249<br>(0.0426)     | -0.1757<br>(0.3381)    | -0.0386<br>(0.0742)    | -0.3798<br>(0.2696)    | -0.0543<br>(0.0384)    | 0.7188<br>(0.7190)     | 0.0715<br>(0.0724)     |
| Advanced Math                    |                       |                       | 1.1482***<br>(0.1999)  |  |                         |                        | -0.0098<br>(0.2366)    | -0.0021<br>(0.0520)    | 0.7178***<br>(0.1795)  | 0.1027***<br>(0.0248)  | -0.7520<br>(0.5452)    | -0.0748<br>(0.0534)    |
| Committee Score                  |                       |                       |                        |  |                         |                        | 0.1714***<br>(0.0604)  | 0.0376***<br>(0.0128)  | -0.0769<br>(0.0684)    | -0.0110<br>(0.0098)    | -1.0032***<br>(0.2212) | -0.0998***<br>(0.0150) |
| Missing Committee Score          |                       |                       |                        |  |                         |                        |                        |                        | 0.1129<br>(0.1909)     | 0.0162<br>(0.0273)     |                        |                        |
| Year Transition/Matriculate More | -0.2523**<br>(0.1282) | -0.0618**<br>(0.0311) | 0.0871<br>(0.2014)     |  | 1.0879***<br>(0.1466)   | 0.2119***<br>(0.0255)  | -0.7286***<br>(0.2324) | -0.1600***<br>(0.0485) | -0.3741**<br>(0.1781)  | -0.0535**<br>(0.0253)  | 5.6048***<br>(0.9335)  | 0.5574***<br>(0.0301)  |
| Program Rank                     |                       |                       |                        |  |                         |                        |                        |                        | -0.0309***<br>(0.0019) | -0.0044***<br>(0.0002) |                        |                        |
| $\sqrt{MSE}$                     |                       |                       | 1.8362***<br>(0.0603)  |  |                         |                        |                        |                        |                        |                        |                        |                        |
| Obs   Param   LL   AIC           | 1000                  | 46                    | -2693.0                |  | 5478.1                  |                        |                        |                        |                        |                        |                        |                        |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

# A4: Simulated Restricted Matriculation-Dynamic Estimates (Near-Optimal)

Table 27a: Simulated Restricted Matriculation-Dynamic Estimates (Near-Optimal)

|                                  | Advanced Math         | Committee Score       | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|-----------------------|-----------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -1.8113**<br>(0.7828) | 1.2405<br>(1.5031)    | -1.4922***<br>(0.1913) | 0.2249<br>(0.1546, 0.3272) | 0.3506<br>(0.2619) | 58.68%<br>(45.94%, 70.35%) | -7.7566***<br>(1.1350) | 2.9160<br>(5.1107)     |                       |
| GRE Quantitative Percentile      | 0.0257***<br>(0.0094) | 0.0408**<br>(0.0173)  |                        |                            |                    |                            | 0.1057***<br>(0.0129)  | -0.0133<br>(0.0562)    |                       |
| Gender                           | -0.2464*<br>(0.1312)  | 0.1731<br>(0.2034)    |                        |                            |                    |                            | 0.1159<br>(0.1353)     | -0.9424*<br>(0.5543)   |                       |
| Demographic 1                    | -0.0487<br>(0.1729)   | -0.7009**<br>(0.2890) |                        |                            |                    |                            | -0.2877<br>(0.1882)    | 1.2129<br>(0.8220)     |                       |
| Demographic 2                    | -0.3098<br>(0.1929)   | -0.4291<br>(0.3258)   |                        |                            |                    |                            | -0.3732*<br>(0.2125)   | 0.5064<br>(0.9746)     |                       |
| Advanced Math                    |                       | 1.1261***<br>(0.2002) |                        |                            |                    |                            | 0.4974***<br>(0.1521)  | 0.3171<br>(0.5299)     |                       |
| Committee Score                  |                       |                       |                        |                            |                    |                            | -0.0238<br>(0.0447)    | -1.0550***<br>(0.2347) |                       |
| Missing Committee Score          |                       |                       |                        |                            |                    |                            | 0.0211<br>(0.1952)     |                        |                       |
| Year Transition/Matriculate More | -0.2441*<br>(0.1282)  | 0.1538<br>(0.2051)    | -1.2324***<br>(0.2881) | 0.0656<br>(0.0355, 0.1212) | 0.4094<br>(0.3147) | 68.14%<br>(57.83%, 76.93%) | -0.3789**<br>(0.1823)  | 4.9595***<br>(0.9084)  |                       |
| Program Rank                     |                       |                       |                        |                            |                    |                            | -0.0301***<br>(0.0018) |                        |                       |
| $\sqrt{\text{MSE}}$              |                       | 1.8440***<br>(0.0615) |                        |                            |                    |                            |                        |                        |                       |
| Sigma                            |                       |                       |                        |                            |                    |                            |                        |                        | 0.1872***<br>(0.0329) |
| Obs   Param   LL   AIC           | 1000                  | 37                    | -2707.6                | 5489.2                     |                    |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

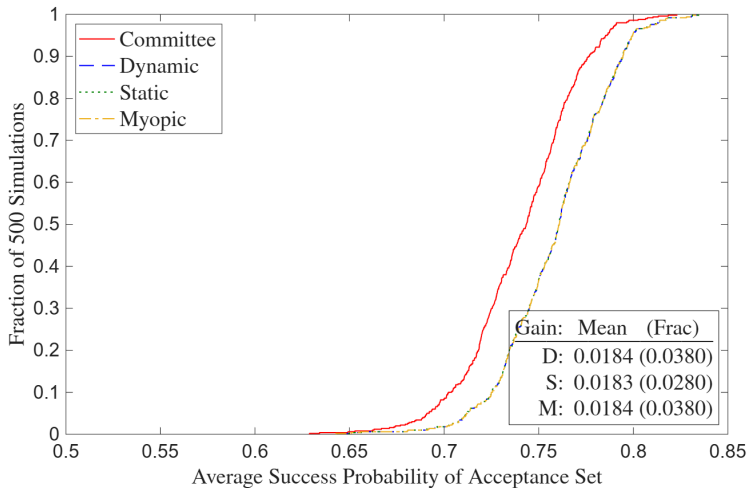
# A4: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

**Table 27b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)**

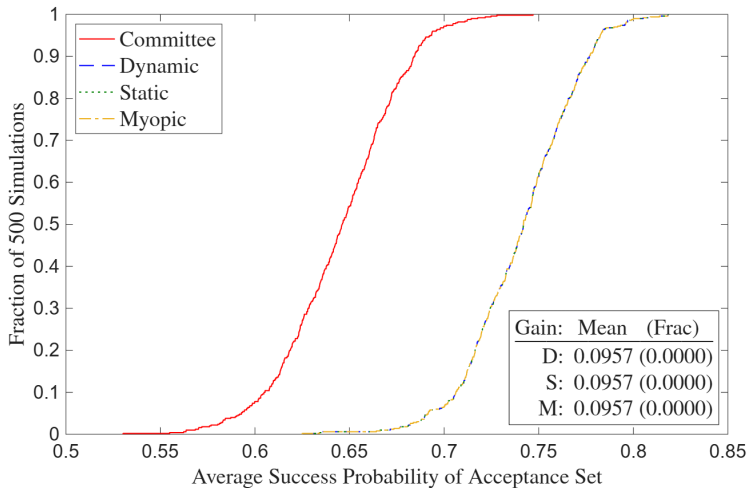
|                                  | Advanced Math         | Committee Score        | Cutoff 1               | $L(\text{Cutoff})_1$       | Cutoff 2           | $P(\text{Cutoff})_2$       | Success                | Matriculate            | Sigma                 |
|----------------------------------|-----------------------|------------------------|------------------------|----------------------------|--------------------|----------------------------|------------------------|------------------------|-----------------------|
| Intercept                        | -1.7155**<br>(0.7813) | 1.8306<br>(1.4228)     | -1.0818***<br>(0.2600) | 0.3390<br>(0.2036, 0.5643) | 0.2612<br>(0.3040) | 56.49%<br>(41.71%, 70.20%) | -7.7219***<br>(1.1701) | 0.2437<br>(4.4569)     |                       |
| GRE Quantitative Percentile      | 0.0246***<br>(0.0094) | 0.0357**<br>(0.0167)   |                        |                            |                    |                            | 0.1020***<br>(0.0133)  | 0.0200<br>(0.0495)     |                       |
| Gender                           | -0.2456*<br>(0.1312)  | 0.1683<br>(0.2035)     |                        |                            |                    |                            | 0.1877<br>(0.1499)     | -1.0474*<br>(0.5402)   |                       |
| Demographic 1                    | -0.0469<br>(0.1731)   | -0.7099***<br>(0.2735) |                        |                            |                    |                            | -0.5368***<br>(0.1983) | 0.5144<br>(0.7157)     |                       |
| Demographic 2                    | -0.3097<br>(0.1929)   | -0.4873*<br>(0.2925)   |                        |                            |                    |                            | -0.3087<br>(0.2284)    | 0.7381<br>(0.7585)     |                       |
| Advanced Math                    |                       | 1.1443***<br>(0.2001)  |                        |                            |                    |                            | 0.5177***<br>(0.1729)  | -0.8396<br>(0.5352)    |                       |
| Committee Score                  |                       |                        |                        |                            |                    |                            | 0.0412<br>(0.0545)     | -0.9473***<br>(0.2114) |                       |
| Missing Committee Score          |                       |                        |                        |                            |                    |                            | 0.0852<br>(0.1912)     |                        |                       |
| Year Transition/Matriculate More | -0.2478*<br>(0.1281)  | 0.0862<br>(0.2022)     | -1.0168***<br>(0.2514) | 0.1226<br>(0.0646, 0.2327) | 0.3707<br>(0.4109) | 65.29%<br>(50.97%, 77.29%) | -0.3780**<br>(0.1794)  | 5.3618***<br>(0.9378)  |                       |
| Program Rank                     |                       |                        |                        |                            |                    |                            | -0.0305***<br>(0.0018) |                        |                       |
| $\sqrt{\text{MSE}}$              |                       | 1.8366***<br>(0.0602)  |                        |                            |                    |                            |                        |                        |                       |
| Sigma                            |                       |                        |                        |                            |                    |                            |                        |                        | 0.2943***<br>(0.0717) |
| Obs   Param   LL   AIC           | 1000                  | 37                     | -2747.6                | 5569.2                     |                    |                            |                        |                        |                       |

Robust standard errors in parentheses: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Confidence intervals are 95%.

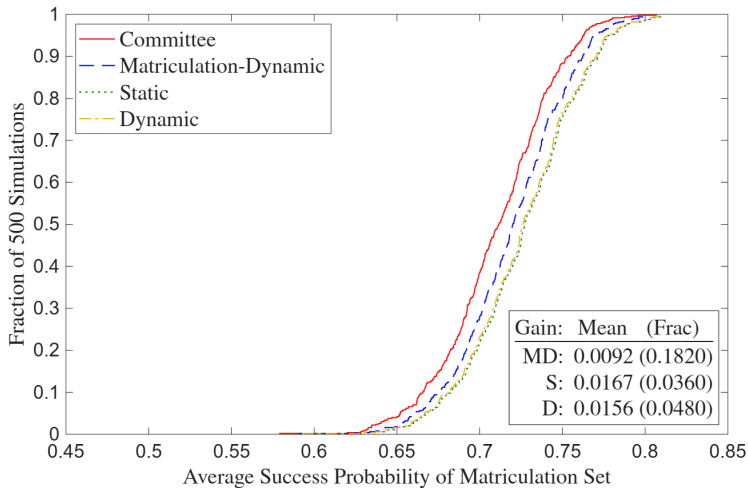
# A4: Simulated Deviation Gains (Near-Optimal)



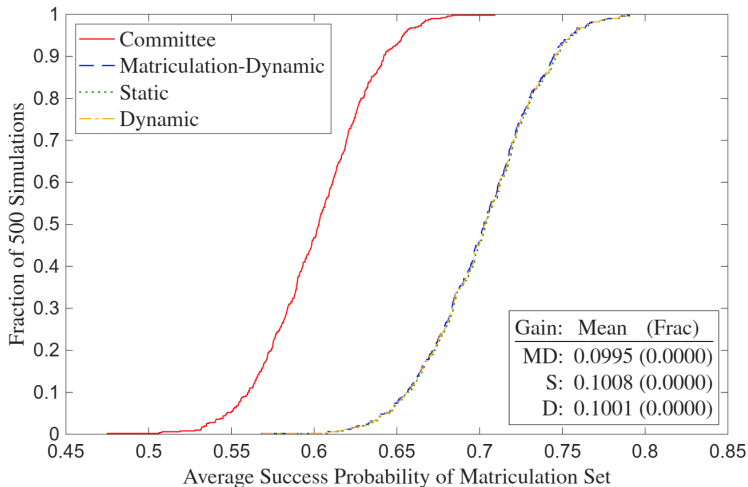
## A4: Simulated Deviation Gains (Suboptimal)



# A4: Simulated Matriculation Deviation Gains (Near-Optimal)



# A4: Simulated Matriculation Deviation Gains (Suboptimal)



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